PHYSICS GRADE 10: GREENHOUSE EFFECT AND CLIMATE CHANGE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 4.1 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 4.0: Environmental Physics
Sub-Strand	Sub-Strand 4.1: Greenhouse Effect and Climate Change
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– The Greenhouse Effect: meaning, mechanism, and key greenhouse gases (CO_2, CH_4, N_2O, water vapour) – Factors leading to the Greenhouse Effect: fossil fuel combustion, deforestation, industrial emissions, agriculture – The Ozone Layer: structure, role in filtering UV radiation, and its relationship to climate – Global Warming: causes, evidence (rising temperatures, melting glaciers, sea-level rise), and effects on the Kenyan environment (drought in the Rift Valley, flooding in Kisumu, reduced rainfall in Kericho tea farms) – Effects of Climate Change on ecosystems, food security (maize and bean harvests), water sources (Lake Victoria, Mt. Kenya glaciers), and human health in Kenya – Mitigation and adaptation measures: renewable energy, reforestation, sustainable agriculture, climate policy (Paris Agreement, Kenya's NDCs) – Appreciating individual and community responsibility in addressing climate change
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the greenhouse effect and how it maintains Earth's temperature, b) identify the major greenhouse gases and describe human activities in Kenya and globally that increase their concentrations, c) describe the role of the ozone layer in protecting life on Earth and explain how its depletion relates to climate change, d) analyse the effects of global warming and climate change on the Kenyan environment, food systems, water bodies, and communities, e) evaluate mitigation and adaptation strategies for addressing climate change at local, national, and global levels, f) appreciate the importance of individual and collective responsibility in responding to climate change.
Core Competencies	– Communication and Collaboration: learners work in groups to research and present findings on the causes and effects of climate change in Kenya, embracing teamwork and contributing to shared decisions during discussions and model-building activities. – Critical Thinking and Problem Solving: learners analyse NASA climate data, PhET simulations, and LabXchange modules to evaluate evidence for global warming, draw conclusions, and propose locally relevant mitigation strategies. – Learning to Learn: learners practise independent inquiry by using internet articles, UNEP data portals, and UN Climate Action resources to deepen their understanding of ozone depletion and climate mitigation, reflecting on how their understanding grows across lessons. – Digital Literacy: learners interact with digital tools — including PhET simulations, LabXchange modules, NASA data portals,

	and YouTube documentaries — to gather, interpret, and present evidence on climate change phenomena. – Citizenship: learners connect global climate science to local Kenyan contexts (Rift Valley droughts, Kisumu floods, Kericho tea-farming disruptions), developing a sense of civic responsibility toward environmental stewardship.
Core Values	– Responsibility: the learner demonstrates personal and community accountability by identifying how everyday choices (energy use, waste disposal, deforestation) contribute to or reduce greenhouse gas emissions in the Kenyan context. – Respect: the learner demonstrates open-mindedness and appreciates diverse opinions when discussing the lived experiences of communities affected by climate change across different regions of Kenya. – Unity: the learner recognises that climate change is a shared challenge requiring collective action — from local farmers in the Rift Valley to international policymakers — and embraces a spirit of cooperation in designing mitigation solutions. – Social Justice: the learner acknowledges that the communities least responsible for greenhouse gas emissions (e.g., rural Kenyan subsistence farmers) often bear the heaviest burden of climate change impacts, fostering empathy and equity-oriented thinking.
Science & Engineering Practices	– Asking Questions and Defining Problems: learners generate and refine questions about why Kenya's climate patterns are shifting — e.g., why is the long rain season becoming unpredictable in Kericho? — to anchor their investigation of the greenhouse effect. – Developing and Using Models: learners build and iteratively revise a physical/conceptual model of the greenhouse effect, adding layers across lessons to show energy absorption, re-radiation, and the role of specific greenhouse gases. – Planning and Carrying Out Investigations: learners engage with PhET and LabXchange simulations as structured investigations, varying greenhouse gas concentrations and observing temperature changes to gather evidence. – Analysing and Interpreting Data: learners examine NASA climate datasets, UNEP ozone data, and local Kenyan meteorological records to identify trends, anomalies, and causal relationships between emissions and observed warming. – Constructing Explanations: learners use gathered evidence to construct written and verbal explanations of the greenhouse effect mechanism, global warming, and ozone depletion, grounding their explanations in data. – Engaging in Argument from Evidence: learners evaluate competing claims about the primary drivers of climate change and defend proposed mitigation strategies using evidence from simulations, data portals, and documentary sources. – Obtaining, Evaluating, and Communicating Information: learners critically assess internet articles, YouTube documentaries, and UN reports for reliability and bias, then communicate synthesised findings through posters, presentations, and the Driving Question Board.
Pertinent & Contemporary Issues (PCIs)	– Environmental Education: the entire sub-strand is rooted in environmental stewardship — learners directly connect the physics of the greenhouse effect to real degradation of Kenyan ecosystems, water bodies, and agricultural land, fostering pro-environmental attitudes and behaviours. – Climate Change Education: learners engage with current, locally relevant climate data (Kenyan NDCs, UNEP reports, NASA records) to understand the urgency of climate action and Kenya's role in global mitigation efforts. – Gender Disparity: learners examine how climate change disproportionately affects women and girls in Kenya (e.g., increased distances to fetch water during droughts, reduced school attendance during floods), and address misconceptions that physics and environmental science are not fields for girls. – Sustainable Development Goals (SDGs): learners explicitly connect their study to SDG 13 (Climate Action), SDG 15 (Life on Land), and SDG 6 (Clean Water and Sanitation), understanding physics as a tool for achieving sustainable development in Kenya.
Career Connections	– Meteorologist / Climate Scientist: understanding the greenhouse effect and climate data analysis is foundational for careers at the Kenya Meteorological Department or international climate research institutions. – Environmental Engineer: knowledge of climate change mitigation informs the design of renewable energy systems, green

	buildings, and sustainable water infrastructure across Kenya. – Agricultural Scientist / Agronomist: understanding shifting rainfall patterns and temperature rises is critical for developing climate-resilient crop varieties and farming practices for Kenyan staple foods like maize, beans, and tea. – Renewable Energy Technologist: the sub-strand's focus on mitigation measures directly connects to careers in solar, wind, and geothermal energy — sectors of major growth in Kenya (e.g., Lake Turkana Wind Power, Olkaria Geothermal). – Policy Analyst / Environmental Lawyer: learners who engage with the Paris Agreement and Kenya's NDCs develop awareness of careers in climate policy, environmental law, and international climate negotiations. – Science Journalist / Educator: skills in evaluating and communicating climate information prepare learners for careers in science communication, documentary production, or teaching environmental science.
Focus for Lessons	This unit investigates the physics of the greenhouse effect and its consequences for climate change, using Kenya's own changing environment — from the shrinking glaciers of Mount Kenya to unpredictable rainfall in Kericho and recurrent floods in Kisumu — as the central context. Learners move from understanding the energy-transfer mechanism of the greenhouse effect to analysing evidence for global warming, evaluating the role of the ozone layer, and designing and defending mitigation strategies. The unit employs a phenomenon-driven, inquiry-based approach in which students build and revise a cumulative model across all seven lessons, anchored by the puzzling observation that Kenyan farmers are experiencing climate shifts that physics can help explain.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable — and what can the physics of our atmosphere tell us about how to respond? KICD KEY INQUIRY QUESTIONS: 1. How does the greenhouse effect influence global and local climates? 2. What are the causes and effects of climate change, and what measures can be taken to address it?
Anchoring Phenomenon	ANCHORING PHENOMENON — "Why Are Mount Kenya's Glaciers Disappearing and Kenyan Farmers Struggling to Predict the Rains?" Students are presented with a striking paired image: a historical photograph of Mount Kenya's Lewis Glacier taken in 1934 alongside a current satellite image showing its dramatic retreat. Alongside this, students read a short testimony from a maize farmer in Embu County who explains that the long rains (March–May) now arrive weeks later than they did in their grandparents' time, and that yields of maize and beans have fallen sharply. Students are asked: "Both of these changes — a shrinking glacier on a volcanic mountain and unpredictable rains on a farm hundreds of kilometres away — are happening at the same time. Could the same cause be responsible for both? What physics might explain this?" This is puzzling because students understand that ice melts when heat is added, but the mountain has not moved closer to the sun. It is also puzzling that the atmosphere — an invisible gas — could be responsible for warming the entire planet and disrupting rainfall patterns over Kenya. The phenomenon is locally grounded, emotionally resonant for Kenyan learners, and scientifically rich enough to sustain seven lessons of inquiry.
Storyline Thread	Lesson 1 — What Is the Greenhouse Effect? (Anchoring Phenomenon Introduction) Students encounter the anchoring phenomenon (Mount Kenya glacier retreat + Embu farmer testimony). They make initial predictions about what could cause both changes simultaneously. Using the "hot car in Nairobi" supporting phenomenon as an analogy, students engage with the PhET Greenhouse Effect simulation to observe how CO_2 and other gases absorb and re-radiate infrared energy. Students begin constructing their initial conceptual model on the Driving Question Board: "What does our atmosphere do with the sun's energy?" Lesson 2 — What Human Activities in Kenya Are Increasing Greenhouse Gases?

	Students investigate the sources of major greenhouse gases (CO_2, CH_4, N_2O) with a specific focus on Kenya: charcoal burning for cooking ugali, livestock (cattle releasing CH_4), deforestation of Mau Forest, and industrial emissions. Using NASA climate data and LabXchange's human activities module, students analyse trends in Kenyan and global emissions. They revise their model to include emission sources, connecting back to the driving question: "Could our daily activities — burning charcoal, clearing forests — be causing the glacier to melt?" Lesson 3 — What Is the Ozone Layer and Why Does It Matter for Kenya? Students investigate the structure and function of the ozone layer using the Khan Academy resource and UNEP Ozone Data Portal. They explore how ozone depletion (caused by CFCs and other chemicals) differs from but interacts with the greenhouse effect, and examine the supporting phenomenon of rising UV-B levels in Nairobi. Students update their model to show the ozone layer as a separate but related atmospheric shield, adding nuance to their understanding of "what the atmosphere does." Lesson 4 — How Is Global Warming Changing Kenya's Environment? Students analyse evidence of global warming's effects using NASA sea-level and temperature data, National Geographic video clips, and the Lake Victoria supporting phenomenon. They map local effects: Mt. Kenya glacier loss, Rift Valley droughts, Kisumu floods, disrupted tea harvests in Kericho. Students build an "effects web" — a visual model linking the greenhouse effect mechanism to cascading environmental and social consequences in Kenya — and return to the farmer's testimony from Lesson 1 to explain it with new scientific language. Lesson 5 — What Can Be Done? Mitigation and Adaptation Strategies Students evaluate a range of mitigation strategies using the UN Climate Action Portal and a YouTube video on renewable energy. They compare Kenya's Olkaria Geothermal expansion and Lake Turkana Wind Farm (national mitigation) with household and community actions (energy-efficient jikos, tree planting, solar panels). Students engage in structured argumentation: which strategies would be most effective and equitable for different Kenyan communities? They update their cumulative model to include feedback loops showing how mitigation actions reduce greenhouse gas concentrations. Lesson 6 — Whose Voices? Appreciating the Human Impact of Climate Change Students watch the LabXchange Climate Change Awareness module and YouTube testimonies from communities affected by climate change (including Kenyan pastoralists in Turkana and coastal communities in Mombasa facing sea-level rise). They analyse how climate change intersects with gender, poverty, and food security in Kenya, connecting to PCIs. Students synthesise their learning by writing a short "letter to a Kenyan policymaker" arguing for a specific climate action, using evidence gathered across all prior lessons. Lesson 7 — Final Model Building, DQB Closure, and Unit Synthesis Students complete their final cumulative model of the greenhouse effect and climate change — from solar energy input through atmospheric trapping, to global warming, to local Kenyan effects, to mitigation pathways. The Driving Question Board is revisited: which questions from Lesson 1 can now be answered? Which new questions have emerged? Student groups present their models and compare them, identifying consensus explanations and remaining uncertainties. The unit closes by returning to the anchoring phenomenon: students write a full scientific explanation of why Mount Kenya's glaciers are retreating and why Embu's farmers are struggling to predict the rains — using every layer of their cumulative model.
Supporting Phenomena	– The Hot Car Phenomenon: a car parked in the Nairobi sun with windows closed becomes dangerously hot inside — far hotter than the air outside — even though sunlight passes through the glass. Learners use this familiar Kenyan experience to build an initial model of how a transparent medium can trap heat, as a concrete analogy for how greenhouse gases trap infrared

	radiation in the atmosphere. – The Ozone Hole Over Antarctica and Rising UV Levels in Nairobi: students examine UNEP data showing that UV-B radiation reaching Nairobi has measurably increased over decades and is linked to higher rates of skin conditions — connecting the physics of ozone depletion to real health consequences in Kenya. – Lake Victoria's Shrinking Shoreline and Changing Fish Catches: fishermen in Kisumu report that the lake's water levels and fish populations are shifting. Students analyse temperature and evaporation data to connect warmer air temperatures (driven by the greenhouse effect) to increased lake evaporation and disrupted aquatic ecosystems — showing that climate change reaches even inland water bodies. – Kenya's Energy Mix Shift — From Thermal to Geothermal and Wind: students observe that Kenya's government has been rapidly expanding the Olkaria Geothermal plant and the Lake Turkana Wind Farm, deliberately moving away from fossil fuel power. This supporting phenomenon drives discussion of mitigation: why are engineers and policymakers making this shift, and how does the physics of the greenhouse effect explain the urgency?

LESSON 1 (80 minutes): What Is the Greenhouse Effect? — Anchoring Phenomenon Introduction	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit's anchoring phenomenon by confronting students with the visible retreat of Mount Kenya's Lewis Glacier and the testimony of an Embu County maize farmer, and to establish that the physics of Earth's atmosphere may explain both observations.
Knowledge	Students will know that (a) the greenhouse effect is the process by which certain atmospheric gases absorb and re-emit infrared radiation, warming Earth's surface; (b) greenhouse gases include water vapour, carbon dioxide, methane, and nitrous oxide; (c) human activities have increased greenhouse gas concentrations since industrialisation; and (d) the Lewis Glacier on Mount Kenya has retreated dramatically since the early 20th century as a measurable indicator of climate change.
Skills	Students will be able to (a) analyse paired historical and contemporary evidence of environmental change; (b) generate scientific questions from an anchoring phenomenon; (c) construct an initial diagrammatic model of how energy enters and is trapped in Earth's atmosphere; and (d) record observations and inferences using a structured T-chart.
Attitudes	Students will appreciate that physics is directly relevant to Kenyan lived experience and that local environmental changes — glacier retreat, shifting long rains — are connected to global atmospheric processes. Students will demonstrate curiosity, open-mindedness about surprising explanations, and a sense of shared responsibility for understanding and responding to climate change.
Key Inquiry Question	How does the greenhouse effect influence global and local climates?
Purpose in Storyline	This is the opening lesson of the unit. It launches the anchoring phenomenon — the disappearing Lewis Glacier and unpredictable Kenyan rains — and opens the Driving Question Board. Students do not yet receive full explanations; instead they surface prior ideas, generate questions, and build an initial (naive) model that will be revised across all seven lessons. Every subsequent lesson will return to this phenomenon and update student understanding.
Safety Notes	No laboratory hazards in this lesson. If printed photographs are used, handle with care. If a candle or lamp is used as an analogy for solar radiation during the model-building phase, the teacher must ensure it is placed on a stable, non-flammable surface, students remain seated, and the teacher controls the flame at all times. Candle use is optional and may be replaced by a torch/LED.

B. LESSON OVERVIEW

Lesson 1 launches the seven-lesson unit on Greenhouse Effect and Climate Change by presenting students with a compelling, locally grounded anchoring phenomenon: the dramatic retreat of Mount Kenya's Lewis Glacier between 1934 and today, paired with a farmer's testimony from Embu County about the failure of predictable long rains. Students first predict possible causes using only prior knowledge, then observe and analyse the paired photographic evidence and farmer testimony, then begin to explain the physics of how an invisible atmospheric gas layer can trap heat — introducing the term 'greenhouse effect' for the first time. The lesson closes with the creation of the class Driving Question Board (DQB), a collaborative anchor that will track the class's growing understanding across all seven lessons, and an initial naïve diagrammatic model of Earth's energy balance that students will revise as the unit progresses. No prior knowledge of greenhouse gases is assumed. NGSS Science and Engineering Practices foregrounded: Asking Questions (Practice 1), Analysing and Interpreting Data (Practice 4), and Developing and Using Models (Practice 2).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown only the title card on the board: 'Mount Kenya's Lewis Glacier — 1934 vs Today' and the farmer quote: 'My grandfather knew exactly when to plant maize. I no longer do.' — James Muriithi, Embu County, 2023. Before seeing any images or data, each student individually writes 3 bullet-point predictions in their exercise books responding to the prompt: 'What do you think could cause a glacier on a high mountain to shrink AND cause rains to become unpredictable on farms far away — at the same time? Could one cause explain both?' Students then share predictions with a partner (Think-Pair-Share). Selected pairs share with the class. Predictions are recorded on a 'Prediction Wall' section of the chalkboard — a two-column T-chart labelled MY PREDICTION | MY REASONING. No predictions are marked right or wrong at this stage.	1. Display the title card and farmer quote. Say exactly: 'Before I show you anything else, I want you to think like a physicist. In your book, write three predictions — what single cause could make a glacier shrink AND make rains unpredictable? You have 4 minutes. Work alone.' [START 4-MINUTE TIMER.] 2. After 4 minutes, say: 'Now share your three predictions with your partner. Decide which one prediction you both find most convincing. You have 2 minutes.' [START 2-MINUTE TIMER.] 3. Cold-call EXACTLY 5 pairs in sequence around the room. For each pair, ask: 'Tell us your best prediction and your reasoning.' Record every prediction verbatim on the Prediction Wall. Do NOT evaluate — say: 'Thank you — that stays on our wall. We will come back to test it.' 4. WAIT TIME: After each pair speaks, wait 10 seconds before calling on the next pair to allow other students to compare their own predictions silently. 5. Before moving on, circle any prediction that mentions 'heat', 'sun', 'air', 'gases', or 'pollution' and say: 'Notice that some of us are already pointing to the atmosphere or heat. Keep that in mind.'	Think-Pair-Share with a Prediction Wall. This strategy activates prior knowledge and creates cognitive dissonance — students hold their prediction publicly on the board and will feel genuine intellectual tension when evidence confirms or disconfirms it. Recording predictions verbatim respects all ideas and models scientific humility. The Prediction Wall functions as a formative baseline for the teacher.	OBSERVABLE INDICATOR: Every student has written at least 3 bullet points in their exercise book before pair-sharing (teacher circulates and visually scans books during the 4-minute window). TEACHER LOOK-FOR: Are any students already proposing atmospheric/heat-related causes? Flag these students — they may serve as intellectual anchors in Phase 3. MISCONCEPTION ALERT: If students say 'the sun is getting closer to Earth' or 'the ozone hole is letting in more sun', note these on the Prediction Wall with a star — they will be directly addressed in Lessons 2 and 3.
Observe Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English)	The teacher now reveals the full anchoring phenomenon in three steps. STEP A — Visual Evidence: Students are shown the paired images: the 1934 photograph of the Lewis Glacier (printed A3 or projected) and the current satellite image showing dramatic retreat.	1. Reveal the paired images and say: 'You now have 5 minutes. In your books, draw the two-column T-chart I am drawing on the board. On the left: write only what you literally see — facts. On the right: write what you think it means — inferences. Go.' [CIRCULATE —	Observation T-Chart (Observation vs Inference). This strategy trains students in a core scientific practice — separating empirical evidence from interpretation — which directly mirrors NGSS Practice 4 (Analysing and Interpreting Data). The data strip	OBSERVABLE INDICATOR: Student T-charts contain at least 2 distinct observations and 2 distinct inferences by the end of Step B. TEACHER LOOK-FOR: Do students distinguish observation from inference, or do they conflate them? Students who write

🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students silently complete an Observation T-chart in their books with columns: WHAT I OBSERVE (facts only) | WHAT I INFER (explanations). They are given 5 minutes. STEP B — Farmer Testimony: Students read aloud (round-robin, one sentence each) the extended testimony of James Muriithi, a maize farmer in Embu County, who describes how the long rains (March–May) now arrive 3–4 weeks later than in his grandfather's era, and how his maize and bean harvests have fallen sharply. Students add new observations and inferences to their T-chart. STEP C — Data Strip: Each student receives a printed data strip showing three data points: Lewis Glacier area in 1934 (~1.35 km²), 2004 (~0.37 km²), and 2023 (estimated <0.20 km²). Students are asked to describe the pattern in one sentence and to estimate when the glacier might disappear entirely if the trend continues. Students share their estimates verbally with the class.	look at books, do not prompt yet.] 2. After 5 minutes, cold-call 3 students — one to share an observation, one to share an inference, one to identify a difference between the two images they found most striking. WAIT TIME: 12 seconds after each question before accepting an answer. 3. Distribute the farmer testimony. Say: 'We will read this together. Each person reads one sentence. Listen carefully — add new entries to your T-chart as we read.' [Conduct round-robin reading.] 4. After the reading, ask: 'James Muriithi's farm is roughly 300 km from Mount Kenya. Why would I show you a glacier AND a farmer's testimony in the same lesson?' [WAIT 15 seconds.] Cold-call 2 students. 5. Distribute the data strip. Say: 'Using only these three numbers, describe the pattern in one sentence and estimate when the glacier will be gone. Write it down, then tell your neighbour.' [2-minute pair task.] Cold-call 3 students for their estimates. Record estimates on the board. 6. Say: 'Hold these numbers in your mind. By the end of this unit, you will be able to explain the physics behind every single one of them.'	adds quantitative reasoning and anchors the phenomenon in measurable change, making it harder to dismiss as anecdotal. The farmer testimony humanises the data and creates emotional resonance for Kenyan learners who may have farming families.	interpretive statements in the 'observation' column need targeted support in scientific reasoning. SECOND INDICATOR: At least one student per group can state the pattern in the glacier data (shrinking over time) and offer a reasonable disappearance estimate. Flag wildly inconsistent estimates for brief whole-class discussion.
Explain Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	The teacher guides students through a structured explanation of the greenhouse effect using an analogy, a diagram, and a short PhET simulation interaction. ANALOGY: Students are first asked to recall what happens inside a car parked in the sun in Nairobi at noon — the interior	1. Begin with the car analogy: 'Raise your hand if you have ever opened a car door after it has been parked in the sun on Moi Avenue in Nairobi. What did you notice?' [WAIT 10 seconds.] Cold-call 3 students. Say: 'The glass let sunlight in but trapped the heat inside. Earth's atmosphere does	Analogy-to-Mechanism Scaffolding. The Nairobi car analogy gives students a concrete, locally familiar mental model before the abstract atmospheric mechanism is introduced. This prevents the greenhouse effect from remaining a vague metaphor — students see the physics	OBSERVABLE INDICATOR: Each student's exercise book contains a correctly labelled four-step greenhouse effect diagram with at least four of the six required labels. TEACHER LOOK-FOR (during circulation): Are students drawing unidirectional arrows only (solar radiation going down)? Prompt:

📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	becomes much hotter than the outside air. The teacher explains that Earth's atmosphere acts similarly: shortwave solar radiation passes through the atmosphere and heats the surface, but the longwave infrared radiation emitted by the warm surface is partially absorbed and re-emitted back downward by greenhouse gases, warming the surface further. DIAGRAM: Students copy and label a teacher-drawn diagram in their books: (1) incoming solar shortwave radiation → passes through atmosphere → absorbed by Earth's surface; (2) Earth's surface emits longwave infrared radiation upward; (3) greenhouse gas molecules in the atmosphere absorb and re-emit this infrared radiation in all directions, including back toward Earth; (4) surface warms further. Students label: solar radiation, infrared radiation, greenhouse gases (CO_2, H_2O, CH_4), absorption, re-emission, and enhanced warming. SIMULATION: If devices are available, students open the PhET 'Greenhouse Effect' simulation and toggle greenhouse gas concentrations between pre-industrial and current levels, observing changes in surface temperature. Students without devices observe the teacher demonstration on a projected screen. Students then discuss: 'How does this explain the glacier and the farmer's rains?'	something similar — but with gases instead of glass. This is the greenhouse effect.' 2. Draw the four-step diagram on the board step by step, narrating each arrow. Say: 'Copy this diagram carefully — you will add to it in every lesson of this unit.' 3. After diagram, ask: 'Which gases act like the glass in our car analogy?' [WAIT 12 seconds.] Accept answers, then write on the board: CO_2 (carbon dioxide), H_2O (water vapour), CH_4 (methane), N_2O (nitrous oxide). Say: 'These are called greenhouse gases. Their concentration in our atmosphere has increased because of human activities — we will explore exactly why in Lesson 2.' 4. Launch PhET simulation (or demonstrate): 'Watch what happens to surface temperature when I increase CO_2 from pre-industrial to today's levels.' Run the simulation. Ask: 'What did you observe?' [WAIT 12 seconds.] Cold-call 2 students. 5. Pose the connection question: 'Now look back at the Lewis Glacier images and James Muriithi's testimony. Using what you just learned about the greenhouse effect, write one sentence in your book that connects the greenhouse effect to what we observed in Phase 2. You have 3 minutes.' [Circulate and read sentences over shoulders.] 6. Cold-call 3 students to share their sentences. Affirm the physics language used. Correct any persistent 'ozone hole' conflation: 'The greenhouse effect and the ozone layer are different phenomena — we will keep them	(radiation trapping) before they encounter the term. The PhET simulation adds dynamic, visual confirmation of the mechanism and engages NGSS Practice 2 (Developing and Using Models). The connection-sentence task requires students to explicitly bridge the new concept back to the anchoring phenomenon, consolidating understanding before the DQB phase.	'Remember — the surface also sends energy upward. Where does your upward arrow go?' SECOND INDICATOR: The three cold-called connection sentences explicitly mention greenhouse gases, infrared radiation, or atmospheric warming as the link between the phenomenon and the mechanism. If fewer than two of three do so, conduct a brief whole-class reteach using the car analogy again before proceeding.

		separate in this unit.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Molecules and Light] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students collaboratively create the class Driving Question Board — a large display that will track growing understanding across all seven lessons. Each student receives two sticky notes (or paper squares if sticky notes are unavailable). On the FIRST sticky note, they write a question that today's lesson has made them want to ask — something they are still confused about or curious about regarding the glacier, the rains, or the greenhouse effect. On the SECOND sticky note, they write one thing they now understand that they did not understand at the start of the lesson. Students bring both notes to a designated section of the wall or board. The teacher (or student volunteers) organises the questions into clusters: (A) Questions about causes of greenhouse gas increase; (B) Questions about how warming links to rainfall; (C) Questions about what can be done; (D) Other physics questions. The class Driving Question — 'Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable — and what can the physics of our atmosphere tell us about how to respond?' — is written in large text at the top of the DQB and remains there for all seven lessons.	1. Distribute two sticky notes/paper squares to each student. Say: 'On the BLUE note — write one question this lesson has made you want to ask. Make it a real question you genuinely don't know the answer to. On the YELLOW note — write one thing you understand now that you did not before today. You have 4 minutes.' [START TIMER. CIRCULATE and prompt any student who is stuck: 'What part of today still feels like a mystery to you?'] 2. Invite all students to bring their notes to the board. Say: 'Place your blue question note first, then your yellow understanding note.' 3. Read aloud 6–8 question notes selected at random. Sort them into clusters A, B, C, D on the board, narrating aloud: 'This question belongs in cluster A — causes. This one belongs in cluster B — rainfall links.' Invite students to suggest alternative groupings. 4. Point to the Driving Question at the top of the DQB and say: 'Every lesson for the next seven weeks will bring us closer to answering this question. At the end of each lesson, we will come back to this board and ask: what can we now explain that we could not before?' 5. Say: 'Your question notes are not graded. There are no wrong questions on this board. A good question is more valuable than a half-understood answer.' 6. Leave the DQB visible for the remainder of the lesson and all subsequent lessons.	Driving Question Board (DQB) — a Project-Based Learning sensemaking anchor. The DQB externalises student thinking, makes the unit's intellectual journey visible, and gives every learner — including less confident speakers — a voice. Clustering questions models how scientists organise inquiry. Returning to the DQB at the end of every lesson creates a cumulative sense of growing understanding, which is strongly motivating. The two-note structure (question + understanding) ensures students articulate both confusion and progress, which are equally important epistemic states.	OBSERVABLE INDICATOR: Every student produces both a question note and an understanding note. TEACHER LOOK-FOR: Question notes that reveal persistent misconceptions (e.g., 'Is the greenhouse effect caused by the ozone hole?') are flagged and placed in a 'MISCONCEPTIONS TO ADDRESS' sub-section of the DQB — these become explicit targets for Lessons 2 and 3. Understanding notes that are vague (e.g., 'I understand climate') prompt a brief one-on-one follow-up from the teacher during the model-building phase: 'Can you be more specific — what is one thing that the atmosphere does that you understand better now?'
Model Building	Students construct their first individual diagrammatic model of	1. Say: 'Now I want you to build your first model of Earth's energy	Progressive Model Revision (NGSS Practice 2 — Developing	OBSERVABLE INDICATOR: Every student's Lesson 1 model

Phase 📹 **VIDEO:** [Video: Molecules and Light](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Earth's energy balance in their exercise books. They are given the following prompt: 'Draw a model that shows: (1) where energy from the sun goes when it reaches Earth; (2) what happens to energy at Earth's surface; (3) how the atmosphere is involved; (4) how your model connects to the Lewis Glacier shrinking and James Muriithi's changing rains.' Students may use arrows, labels, colour (if coloured pencils are available), and short explanatory sentences alongside their diagram. Students are explicitly told: 'This is your LESSON 1 model. It does not need to be correct — it needs to be honest. Draw what you currently believe. You will improve this exact model in every lesson that follows.' After drawing individually (7 minutes), students share their model with a partner and identify one thing their partner's model shows that theirs does not. One or two volunteers share their model with the class under the document camera or by holding it up.	system. This is not a test. It is a snapshot of your thinking today. Open to a full blank page in your book and follow the four-point prompt I am writing on the board.' [Write the four prompt points.] 2. [START 7-MINUTE TIMER. CIRCULATE continuously. Do not correct models — instead ask probing questions: 'Where does the infrared radiation go in your model?' / 'You've drawn the sun — what happens after sunlight hits the Aberdare ranges?' / 'Can you add the greenhouse gases somewhere?'] 3. After 7 minutes: 'Show your model to your partner. Tell them one thing you are not sure about in your own model. Then identify one feature in their model that yours is missing. You have 2 minutes.' 4. Cold-call 2 volunteers to share their model. For each, ask the class: 'What does this model explain well?' [WAIT 10 seconds.] Then: 'What is one thing missing or that we might improve later?' [WAIT 10 seconds.] Record suggestions on the board. 5. Say: 'Write today's date and the label LESSON 1 MODEL at the top of your diagram. Do not erase it — ever. In Lesson 7 you will compare this model to your final model and see how your thinking has grown.' 6. Close by returning to the Driving Question: 'Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable? You cannot fully answer that yet. But look at what you have built today — you have already started.'	and Using Models). Requiring students to date and preserve their Lesson 1 model creates a longitudinal record of conceptual change — one of the most powerful metacognitive tools in science learning. The four-part prompt ensures the model is explicitly connected to the anchoring phenomenon from the very first iteration, not treated as an abstract textbook diagram. Partner sharing with a 'one thing missing' protocol encourages peer sensemaking without competitive evaluation.	contains at least (a) the sun as an energy source, (b) Earth's surface, (c) the atmosphere (even if labelled vaguely as 'air'), and (d) at least one arrow showing energy movement. TEACHER LOOK-FOR: Models that show energy flowing only downward (sun → Earth) with no upward re-emission indicate a significant gap that is the explicit target of Lesson 2. Models that include greenhouse gases by name indicate stronger prior knowledge — these students may be assigned peer-teaching roles in Lesson 2. MISCONCEPTION TO NOTE: Any model showing the sun 'moving closer' or a 'hole in the sky' is flagged for Lesson 3 reteach.

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions before planning Lesson 2: 1. PREDICTION WALL — What were the most common student predictions? Did any students already propose atmospheric/gas-related causes? How will you leverage those students as intellectual anchors in Lesson 2? 2. MISCONCEPTIONS IDENTIFIED — Did students conflate the greenhouse effect with the ozone hole, or suggest the sun is 'moving closer'? List these explicitly. Which ones appeared on the DQB question notes? Plan targeted corrections for Lesson 2's explain phase. 3. DQB QUALITY — Are the student questions specific enough to drive inquiry, or are they too vague (e.g., 'Why is the weather changing?')? If questions are too vague, model more precise scientific question-writing at the start of Lesson 2. 4. MODEL BASELINE — What proportion of students produced models with bidirectional energy flow (including upward infrared re-emission)? If fewer than 30% did so, begin Lesson 2 with an extended radiation analogy before introducing the electromagnetic spectrum. 5. ENGAGEMENT — Did the farmer testimony resonate? Did any students mention personal or family experience with changing rain patterns in their own counties (Kisii, Meru, Trans Nzoia, Nakuru)? These testimonies can be integrated into Lesson 3's climate data analysis. Record their names. 6. TIMING — Which phase ran longest? If Phase 3 (Explain) exceeded 20 minutes, consider moving the PhET simulation to the opening of Lesson 2 and using only the car analogy and diagram in Lesson 1.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students examined paired images of Mount Kenya's Lewis Glacier (1934 vs. present day) showing significant ice retreat, read the testimony of James Muriithi, a maize farmer from Embu County, describing the failure of predictable long rains, and analysed a data strip showing the glacier's area shrinking from approximately 1.35 km² in 1934 to less than 0.20 km² in 2023.
What did I learn?	Students learned that the greenhouse effect is the process by which greenhouse gases (carbon dioxide, water vapour, methane, nitrous oxide) in Earth's atmosphere absorb longwave infrared radiation emitted by Earth's surface and re-emit it back toward the surface, causing additional warming. They learned that this process — amplified by human activity — is the underlying physics that connects glacier retreat on Mount Kenya to disrupted rainfall patterns experienced by Kenyan farmers.
How does this explain the phenomenon?	Students began to explain that both the retreat of Mount Kenya's Lewis Glacier and the unpredictability of the long rains in Embu County can be linked to a common cause: enhanced warming of Earth's surface and lower atmosphere driven by increased concentrations of greenhouse gases. Students constructed an initial diagrammatic model of Earth's energy balance and generated questions for the class Driving Question Board that will be investigated across the remaining six lessons of this unit.

LESSON 2 (40 minutes): What Human Activities in Kenya Are Increasing Greenhouse Gases?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the specific human activities in Kenya that are increasing concentrations of greenhouse gases in the atmosphere, and to connect those activities to the glacier retreat and rainfall disruption introduced in Lesson 1.
Knowledge	Students will be able to identify the major sources of CO_2, CH_4, and N_2O in Kenya (charcoal burning, livestock, deforestation, industrial activity); explain how these activities increase atmospheric greenhouse gas concentrations; and describe trends in Kenyan and global emissions using data.
Skills	Students will practise analysing emissions data from NASA and LabXchange, interpreting bar graphs and trend lines, constructing evidence-based claims, and revising a scientific model in light of new evidence.
Attitudes	Students will develop a sense of personal and community relevance — recognising that familiar daily activities such as cooking ugali on a charcoal jiko are connected to large-scale atmospheric change — without fostering guilt but instead cultivating informed agency.
Key Inquiry Question	What are the causes and effects of climate change, and what measures can be taken to address it?
Purpose in Storyline	Lesson 1 established the physics mechanism of the greenhouse effect. Lesson 2 now asks: where are the extra greenhouse gases coming from, and are Kenyan activities contributing? This is the evidence layer that links human behaviour to the atmospheric physics students already understand, and it directly advances the driving question by connecting daily Kenyan life — burning charcoal, raising cattle, clearing the Mau Forest — to the glacier loss and rainfall disruption of the anchoring phenomenon.
Safety Notes	No hazardous materials are used. If discussing deforestation or farming practices, be sensitive to students whose families depend on charcoal trade or subsistence farming. Frame the lesson around systemic causes and shared responsibility rather than individual blame.

B. LESSON OVERVIEW
Students begin by predicting which everyday Kenyan activities might release greenhouse gases, drawing on their own lived experience. They then observe emissions data for Kenya and the world using a printed NASA global emissions infographic and a guided walkthrough of the LabXchange Human Activities and Greenhouse Gases module (or a teacher-led screen projection if internet is limited). In the Explain phase, students connect the data to the physics from Lesson 1, constructing evidence-based claims about how charcoal burning, cattle farming, and Mau Forest deforestation increase CO_2 and CH_4 concentrations. They update the Driving Question Board and revise their cumulative atmospheric model to include emission sources feeding into the greenhouse gas layer.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Video: Molecules and Light] Source: PhET	Students are shown a single slide or poster with four images arranged in a grid: (1) a woman cooking ugali on a charcoal jiko in Nairobi, (2) a herd of cattle on a	Display the four-image grid prominently. Say: We know from last lesson that greenhouse gases trap heat in our atmosphere. Today I want you to think like a	Elicit-before-explain approach: activating prior knowledge and lived experience before presenting data prevents passive reception. The familiar Kenyan scenes (jiko,	Teacher scans sticky notes during the 3-minute pair discussion, listening for: (a) whether students can name any specific gas (CO_2, CH_4), (b) whether they include

Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Renewable Resources and Alternative Energy Sources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	dry Rift Valley plain, (3) a satellite image of the Mau Forest showing deforested patches, and (4) a cement factory chimney near Athi River. The teacher asks students to work in pairs for 3 minutes and write on sticky notes: Which of these activities do you think adds greenhouse gases to the atmosphere? Which gas do you think each activity releases? Students place their sticky notes in a two-column table drawn on the board labelled Activity and Gas Released. The teacher does NOT correct predictions yet — all ideas are valued and posted visibly.	Kenyan scientist. Look at these four scenes from our country. In pairs, decide: which of these activities — if any — adds greenhouse gases to the air? Write your prediction on a sticky note and bring it up. WAIT 3 MINUTES before collecting. After students post, say: I can see some really interesting predictions up here. Some of you think cattle release greenhouse gases — that is a fascinating idea we will investigate. Let us keep these predictions visible so we can come back and check them.	cattle, Mau Forest, Athi River factory) make the scientific question personally relevant and lower the entry barrier for all learners.	deforestation as a source (often overlooked), and (c) whether they show misconceptions such as believing only factories produce greenhouse gases. These inform the depth of explanation needed in the Observe phase.
Observe Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Renewable Resources and Alternative Energy Sources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive a one-page printed data sheet containing: (1) a bar graph of Kenya sectoral greenhouse gas emissions (agriculture, energy, land use, waste, industrial processes) adapted from Kenya National Climate Change Action Plan data, and (2) a simplified global pie chart of GHG sources from NASA Earth Observatory. The teacher projects or displays the LabXchange Human Activities and Greenhouse Gases interactive (or a printed version if offline), walking students through how each source category releases specific gases. Students work in groups of three to complete a structured observation table with columns: Source, Gas Released, Percentage of Kenyan Emissions, and One Surprising Finding. Groups have 8 minutes to read the data sheet and complete the table.	Distribute the data sheet. Say: You have 8 minutes to study these two graphs with your group. I want you to fill in the observation table. Pay special attention to the agriculture and land-use sectors for Kenya — you might be surprised by what you find. WAIT 2 MINUTES for groups to work before circulating. While circulating, ask targeted questions: What does the agriculture bar tell you about farming and greenhouse gases? What gas do cattle release, and where does it come from in a cow? What happens to CO2 storage when a forest is cut down? If projecting LabXchange, pause on the livestock and deforestation slides and say: Notice that in Kenya, agriculture and land-use change together account for more than 75 percent of our national emissions. How does that compare to what you predicted?	Data-first observation before teacher explanation requires students to construct meaning from evidence rather than receive conclusions. The structured observation table scaffolds disciplined noticing. Connecting to the LabXchange module links the local Kenya data to a global scientific framework, showing students that Kenyan scientists and global researchers are looking at the same phenomenon.	Teacher checks observation tables during circulation. Look for: (a) correct identification of CH4 from livestock enteric fermentation, (b) understanding that deforestation releases stored CO2, (c) recognition that agriculture — not factories — dominates Kenyan emissions. Common error: students may assume industrial emissions dominate because factories are visually dramatic. If this appears, ask: Compare the agriculture bar to the industrial bar. Which is taller for Kenya?
Explain Phase 📹 **VIDEO:** Video: Molecules and Light	Each group shares one finding from their observation table in a whole-class share-out (30 seconds per group). The teacher uses	Facilitate the share-out by calling on each group. After all groups share, say: Look at what we have built together. Charcoal burning —	The sentence frame makes the structure of a scientific claim explicit and accessible. The share-out creates collaborative sense-	Collect or photograph student evidence claims. Assess for: (a) correct identification of source, gas, and mechanism, (b) accurate

Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Renewable Resources and Alternative Energy Sources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	student responses to build a class summary on the board in a T-chart: Source on the left, Gas and Mechanism on the right. Students then individually write a 3-sentence evidence claim in their notebooks using the sentence frame: In Kenya, [source] releases [gas] because [mechanism]. This increases the greenhouse effect by [connection to Lesson 1 physics]. For example: In Kenya, burning charcoal releases CO_2 because carbon in the wood combines with oxygen during combustion. This increases the greenhouse effect by adding more CO_2 molecules to the atmosphere, which absorb more infrared radiation re-emitted by Earth's surface, trapping more heat. Students then return to their Lesson 1 predictions on the DQB and check which questions this new data helps to answer.	something millions of Kenyan families do every day to cook ugali — releases CO_2. Cattle farming — central to Maasai and Samburu livelihoods and to beef and milk production for all of us — releases methane. Deforestation of the Mau Forest releases the stored carbon those trees spent decades absorbing. Now I want you to write your evidence claim individually. WAIT 4 MINUTES for writing. Then say: Who would like to read their claim? WAIT 10 SECONDS for volunteers. After one or two students share, ask the class: Does this claim connect back to the physics from last lesson? Can someone point to where the mechanism matches what we saw in the PhET simulation? This explicit connection back to Lesson 1 is essential — the physics mechanism and the source of gases must be linked in students minds.	making from data. The deliberate callback to the PhET simulation from Lesson 1 helps students integrate new evidence into their existing mental model rather than storing it as an isolated fact.	connection to the infrared absorption physics from Lesson 1, (c) use of precise language (infrared radiation, concentration, absorb, re-emit). Provide verbal feedback during the 4-minute writing window by circulating and commenting quietly on student notebooks.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Renewable Resources and Alternative Energy Sources (Flexbook) Source: CK-12 🔍 Search ARES for similar	The teacher returns to the Driving Question Board from Lesson 1, which displays students initial questions and the anchoring phenomenon images. Students receive two new sticky note colours: green for This lesson helps us answer this question and orange for This lesson raises a new question I did not think of before. In pairs, students spend 3 minutes reviewing the DQB and placing their green and orange notes. New questions that typically emerge: Does Kenya produce enough emissions to matter globally? Could we reduce emissions without hurting farmers and herders? Are there other countries causing more damage than us? The teacher clusters new	Point to the DQB and say: Last lesson, you asked big questions about our atmosphere and why things are changing. Today we collected new evidence. Take two sticky notes — green means this lesson helped answer a question already on the board, orange means today raised a brand new question for you. WAIT 3 MINUTES for pairs to write and post. Then step back and read aloud two or three green notes: It sounds like several of you now feel you can explain why charcoal burning could be connected to the glacier. That is real scientific progress. Read aloud one or two orange notes and say: These new questions are exactly what scientists ask. We will come back	The DQB is a living document of the classes inquiry journey. Using two sticky note colours makes progress visible and honours new uncertainty simultaneously — both are valued in science. Clustering orange notes shows students that their questions are coherent and will be systematically addressed, building trust in the inquiry structure.	Green notes reveal which learning goals students feel they have met — a quick proxy for self-assessed understanding. Orange notes reveal remaining gaps and misconceptions. If many orange notes cluster around whether Kenya matters globally, this signals a need to address scale and cumulative emissions explicitly in Lesson 4.

readings	orange notes into emerging themes on the DQB without resolving them — these will drive Lessons 3 through 5.	to them in Lessons 3, 4, and 5. Do not lose that curiosity.		
Model Building Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Renewable Resources and Alternative Energy Sources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to their individual cumulative atmospheric model begun in Lesson 1. The Lesson 1 model showed: solar energy entering the atmosphere, greenhouse gases absorbing and re-emitting infrared radiation, and a warming Earth surface. Students now add a new layer to their model: arrows and labels showing emission sources feeding into the greenhouse gas layer. They must include at least three Kenyan sources (charcoal burning, livestock, Mau Forest deforestation) with the specific gas each releases (CO2, CH4, N2O). They annotate each arrow with a brief mechanism phrase. Students also add a simple trend indicator — a small upward arrow labelled increasing concentration — to show that human activities are raising the amount of greenhouse gas in the atmosphere above the pre-industrial baseline. At the end of 5 minutes, one volunteer shares their updated model under the document camera or by holding it up, and the class gives one piece of warm feedback (something they notice is strong) and one piece of cool feedback (something they would add or change).	Say: Open your model from Lesson 1. You are going to add the sources layer. You have 5 minutes. Your model must show where the greenhouse gases are actually coming from — not just that they exist. Include at least three Kenyan sources. WAIT 5 MINUTES, circulating to check for misconceptions. Common issue: students draw factories as the only source. Prompt: Your data sheet showed Kenya emissions — what was the largest sector? After the volunteer shares, facilitate feedback by asking: What is one thing you see in this model that correctly shows the science? WAIT 5 SECONDS. Then: What is one thing you would add to make it even more complete? WAIT 5 SECONDS. Close by saying: By the end of Lesson 7, this model will show the full story — from human activities all the way to the glacier on Mount Kenya and the rains in Embu. You are building that story piece by piece.	Iterative model revision is a core science practice. Adding a sources layer makes the model mechanistically complete — gases do not appear from nowhere, they come from identifiable human activities. The peer feedback protocol (warm and cool) builds scientific communication skills and models the collaborative norms of real science communities.	Check updated models for: (a) presence of at least three distinct Kenyan sources, (b) correct gas matched to each source (CO2 from burning and deforestation, CH4 from livestock), (c) arrows showing the pathway from source into the atmospheric greenhouse gas layer, (d) upward trend indicator. If students are only drawing factories, use targeted questioning to redirect to the data showing agriculture and land use dominance in Kenya.

D. TEACHER REFLECTION

After the lesson, consider: Did students make the connection between familiar daily Kenyan activities and greenhouse gas emissions, or did they still view emissions as primarily an industrial or foreign problem? Were students able to link the new source data back to the infrared absorption physics from Lesson 1, or did those two ideas remain disconnected? Did the evidence claims show mechanistic thinking (because and by connecting source to gas to physics) or surface-level association? Are there students whose families depend on charcoal burning or cattle farming who appeared uncomfortable — and if so, how can framing in Lessons 3 through 5 better honour

the complexity of livelihood versus environmental trade-offs? What new questions on the DQB need to be addressed earlier than planned?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed NASA and Kenya emissions data showing that agriculture and land-use change (including deforestation) are responsible for more than 75 percent of Kenya total greenhouse gas emissions, and that burning charcoal, cattle farming (releasing methane through digestion), and clearing the Mau Forest are among the largest national sources.
What did I learn?	We learned that CO_2 is released by burning charcoal and by deforestation (when stored carbon in trees is released), that CH_4 is released by cattle through enteric fermentation, and that N_2O is released by agricultural soils. These activities increase the concentration of greenhouse gases in the atmosphere above natural levels.
How does this explain the phenomenon?	We explained that increasing the concentration of greenhouse gases in the atmosphere — through Kenyan and global human activities — means more infrared radiation emitted by the Earth surface is absorbed and re-emitted back toward the surface, which intensifies the greenhouse effect. This is why the glacier on Mount Kenya is retreating and why climate patterns affecting Embu farmers are changing: the same physics mechanism identified in Lesson 1 is being amplified by human activities happening right here in Kenya every day.

LESSON 3 (80 minutes): How Does the Greenhouse Effect Actually Work? — Tracing Energy Through Kenya's Atmosphere	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students construct a mechanistic explanation of the greenhouse effect by tracing the absorption and re-emission of infrared radiation by greenhouse gas molecules, connecting this energy-transfer model to the observed melting of Mount Kenya's Lewis Glacier and shifting rainfall patterns in Embu County.
Knowledge	By the end of this lesson, the learner should be able to: (a) describe how incoming solar shortwave radiation passes through the atmosphere and is absorbed by Earth's surface; (b) explain how the Earth's surface re-emits energy as longwave infrared (IR) radiation; (c) explain how greenhouse gas molecules (CO_2, H_2O vapour, CH_4) absorb and re-emit IR radiation, trapping heat in the lower atmosphere; (d) distinguish between the natural greenhouse effect (essential for life) and the enhanced greenhouse effect caused by rising GHG concentrations.
Skills	The learner is able to: analyse a PhET 'Greenhouse Effect' simulation to identify variables; construct an annotated energy-flow diagram showing shortwave and longwave radiation pathways; use evidence from the simulation and a data table to argue that increasing CO_2 concentration raises equilibrium surface temperature.
Attitudes	The learner appreciates that the physics of electromagnetic radiation — invisible processes occurring in the atmosphere — has direct, visible consequences for Kenyan landscapes and farming communities, and approaches climate evidence with scientific rigour rather than fatalism.
Key Inquiry Question	How does the greenhouse effect influence global and local climates?
Purpose in Storyline	Lessons 1 and 2 established WHAT is changing (retreating glaciers, unpredictable rains) and WHICH human activities in Kenya are emitting greenhouse gases. Lesson 3 now addresses the missing mechanistic link: HOW do those gases physically trap energy in the atmosphere? This is the conceptual engine of the entire driving question. By the end of this lesson, students can explain — at the particle and radiation level — why adding more CO_2 to the air above Kenya raises surface temperatures, bringing them one causal step closer to answering the anchoring phenomenon.
Safety Notes	No hazardous materials are used in this lesson. Students interact with a digital PhET simulation and paper-based materials only. Ensure screen brightness is comfortable for all learners. If the school has limited devices, arrange pairs or groups of three at shared screens. Ensure extension cables and power boards do not create trip hazards.

B. LESSON OVERVIEW

This 80-minute lesson is the mechanistic core of the 7-lesson sequence. Students begin by predicting how the atmosphere traps heat using only their prior knowledge from Lessons 1–2, then investigate the PhET 'Greenhouse Effect' simulation (or a teacher-led guided walkthrough on a projector if devices are limited) to observe how changing CO_2 and CH_4 concentrations alter the infrared energy balance. In the Explain phase, students construct a labelled energy-flow diagram — connecting shortwave solar input, surface absorption, infrared re-emission, and GHG absorption/re-emission — and use it to explain WHY Mount Kenya's glaciers are melting and WHY rainfall patterns over the Rift Valley are shifting. The lesson closes with students updating the class Driving Question Board and adding a new layer to their cumulative atmospheric model. Kenyan contexts (charcoal burning in Kibera, cattle farming in Laikipia, maize growing in Trans-Nzoia) are woven throughout to maintain local relevance.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students individually read a 4-sentence stimulus card: 'The average temperature of the Moon's surface facing the Sun is about 127 °C during the day, but −173 °C at night. The Moon is almost the same distance from the Sun as the Earth. Yet Earth's average surface temperature stays around 15 °C day and night. Earth has an atmosphere. The Moon does not.' Students then write a 3-sentence prediction in their science notebooks answering: 'What role does Earth's atmosphere play in keeping our planet warm enough for life — and could that same role explain why Mount Kenya's glaciers are melting?' Students also sketch a simple arrow diagram showing what they think happens to sunlight after it reaches Earth. Pairs share predictions with each other before a class discussion.	Distribute stimulus cards. Say: 'Read this silently for 90 seconds. Then write your prediction — three sentences minimum. You have four minutes.' WAIT in silence for 4 minutes — circulate and scan notebooks without commenting yet. After writing, say: 'Turn to your partner. Share your diagram. You have 90 seconds each.' After pair-share, cold-call 4 students (aim for gender balance and include at least one student who drew an arrow diagram that shows sunlight bouncing straight back to space — a common misconception to surface). Ask: 'Amina, walk us through your diagram — where does the energy go after it hits the ground?' WAIT 12 seconds after each student responds before prompting or moving on. Record key ideas and misconceptions (e.g., 'the ozone layer keeps us warm', 'clouds trap heat') on the left side of the whiteboard under the heading 'Our Current Ideas'. Do NOT correct misconceptions yet — they will be revisited in Phase 3.	Predict-then-Explain Routine: By committing to a prediction in writing before instruction, students activate prior knowledge, expose pre-existing mental models, and create a cognitive 'knowledge gap' that motivates the observation phase. Surfacing the Moon-vs-Earth temperature anomaly creates productive dissonance — students sense their current model is incomplete, which primes them to look carefully at the simulation evidence.	Teacher scans notebook diagrams during circulation. Observable indicator: Does the student's diagram show energy doing SOMETHING after hitting the surface (even if incorrect), or does it simply show sunlight arriving? Students who draw no surface interaction need targeted prompting in Phase 2. Teacher notes on a clipboard which students hold the 'ozone layer = greenhouse effect' misconception (a common conflation) for deliberate cold-calling during Phase 3.
Observe Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure	Students open the PhET 'Greenhouse Effect' simulation (https://phet.colorado.edu/en/simulations/greenhouse-effect) — either individually, in pairs, or as a teacher-guided whole-class walkthrough on a projector, depending on device availability. Students work through a structured investigation guide (printed worksheet) with four steps: (1) Set the atmosphere to 'No Atmosphere' — observe and	Before starting: 'We are about to use a physics simulation built by scientists at the University of Colorado. Your job is to be a careful observer — like Dr. Wangari Maathai observing her forests, you are looking for patterns in data.' Display the investigation guide on the projector. Walk through Step 1 together as a class: 'Set it to No Atmosphere. Watch for 20 seconds. What is the surface	Structured Inquiry with Data Tables: The stepwise simulation protocol ensures students vary one factor at a time (GHG concentration) while holding others constant, mirroring authentic scientific methodology (NGSS Science & Engineering Practice 3: Planning and Carrying Out Investigations). The colour-coded arrow diagram converts an abstract photon-energy process into a spatial, visual representation	Teacher checks the two-column data table during circulation. Observable indicator: Has the student recorded at least 3 temperature readings with correctly increasing values as GHG concentration increases? Observable indicator for diagram: Are yellow (shortwave) arrows shown passing THROUGH GHG molecules unimpeded, while red IR arrows are shown being absorbed and re-emitted (not

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	record the surface temperature at equilibrium. (2) Set atmosphere to 'Today's Atmosphere' — observe how IR photons behave differently from shortwave photons; record equilibrium temperature. (3) Increase greenhouse gas concentration to '1750 (pre-industrial)' then to 'Ice Age' — record temperatures and sketch the photon behaviour for each. (4) Switch to the 'Photon Absorption' tab — select CO_2, then CH_4, then N_2 — fire one IR photon at each and observe which molecules absorb it. Students record all data in a two-column table: 'Atmosphere Condition | Equilibrium Surface Temperature (°C)'. They also annotate a printed blank atmosphere diagram with coloured arrows: yellow for incoming shortwave, red for outgoing IR, and orange curved arrows for re-emitted IR from GHG molecules. At the end, students answer on their worksheet: 'What pattern do you notice between GHG concentration and surface temperature? Write one sentence.'	temperature? Everyone record that number NOW.' Cold-call 2 students to confirm the reading. Then say: 'Now work through Steps 2, 3, and 4 with your partner. You have 18 minutes. I will call time at 10 minutes — by then you must have finished Steps 1 and 2.' CIRCULATE actively: stop at each table for 30–45 seconds. Ask: 'Show me on your diagram — where is the IR photon going after it hits the CO_2 molecule?' and 'Why did the temperature CHANGE when you added more CO_2 — what did you actually SEE happening differently?' At the 10-minute mark, pause the class: 'Hands up if your temperature went UP when you added more greenhouse gases. Good. Hands up if a CO_2 molecule absorbed the IR photon. Good. Hands up if a nitrogen molecule absorbed it.' WAIT 8 seconds on the last question — it should be silent (N_2 does not absorb IR). Say: 'Notice that silence. We will explain WHY in Phase 3.' After 18 minutes, say: 'Write your one-sentence pattern statement now. You have 90 seconds.'	that students can reason with. The 'Photon Absorption' tab provides a molecular-level observation that directly challenges the misconception that 'all gases trap heat equally'.	passed through)? Students who show IR arrows passing straight through need targeted re-teaching using the 'Photon Absorption' tab. Collect 5 random worksheets at the end of Phase 2 for rapid scanning before Phase 3.
Explain Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES	Teacher delivers a focused 12-minute direct instruction segment using a projected annotated diagram (or whiteboard drawing) that formalises the energy-flow model students observed. Key concepts introduced with precise physics language: (a) Incoming solar radiation is predominantly shortwave (visible + UV); (b) Earth's surface absorbs this and re-emits it as longwave infrared radiation — this is Kirchhoff's/Stefan–Boltzmann's principle at an accessible level; (c)	Open with: 'Let's go back to our prediction wall. [Point to the whiteboard.] Someone predicted that clouds trap heat. Someone predicted the ozone layer keeps us warm. Let us see how our simulation evidence helps us refine those ideas.' Cold-call the student who held the ozone misconception: 'Baraka, in the simulation, which molecules absorbed the infrared photon — was it ozone, or was it CO_2 and CH_4?' WAIT 12 seconds. After student responds, affirm the	Explain-It-to-a-Stranger Task (also known as the Feynman Technique): Writing for a non-expert audience (the Embu farmer) forces students to translate formal physics vocabulary into causal narrative, revealing gaps in their own understanding. The peer 'glow and grow' protocol builds communication and collaboration competencies while providing the teacher with a second wave of formative data. Anchoring the explanation to the phenomenon (the farmer's testimony) maintains	Teacher reads 'Explain It to a Farmer' paragraphs while students do peer feedback. Observable indicator: Does the student's explanation include a causal chain (human activity → more GHGs → more IR absorbed → more back-radiation → higher surface temperature → glacier melt / disrupted evaporation → unpredictable rain)? A complete causal chain with all 5 required vocabulary terms = mastery. Missing or broken causal links = needs re-teaching. Teacher flags

for similar readings	GHG molecules (CO_2, H_2O, CH_4) have molecular bond geometries that allow them to absorb IR photons and re-emit them in all directions, including back toward the surface; (d) This 'back-radiation' raises equilibrium surface temperature above what it would be with no atmosphere — the natural greenhouse effect; (e) Adding more GHGs (from burning charcoal in Nairobi, from cattle on Laikipia ranches, from clearing forest in Mau) increases back-radiation, raising temperatures beyond natural levels — the enhanced greenhouse effect. Students then complete an 'Explain It to a Farmer' task: they write 4–5 sentences explaining to the Embu maize farmer from the anchoring phenomenon exactly why burning more charcoal and clearing more trees is connected to Mount Kenya's glaciers melting. They must use the terms: shortwave radiation, infrared radiation, greenhouse gas, re-emission, and surface temperature. Pairs read each other's explanations and provide one 'glow' (something well explained) and one 'grow' (something to improve).	correct element and clarify the distinction: 'The ozone layer absorbs ULTRAVIOLET radiation from the Sun — that is a different and very important job. The greenhouse effect involves INFRARED radiation being trapped by CO_2 and methane. Two different gases, two different radiation types, two very important processes.' During the direct instruction, pause at each key concept and ask a targeted question: 'Why does the shortwave radiation pass THROUGH the CO_2 molecule without being absorbed — what does that tell us about the relationship between the molecule and the wavelength of the radiation?' WAIT 15 seconds before cold-calling. After the 'Explain It to a Farmer' task, cold-call 3 students to read their explanations aloud. After each: 'Class, thumbs up if they used the term infrared radiation correctly. What would you add to make it even clearer for the farmer?' Record the best class-generated sentence on the board under 'Our Revised Understanding'.	the storyline thread and reinforces why this physics content matters.	2–3 strong exemplars to share with the class at the start of Lesson 4.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	Students return to the class Driving Question Board displayed at the front of the room. The DQB has been running since Lesson 1 with the anchoring question: 'Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable?' Each student receives two sticky notes: one green ('Evidence I can now explain') and one yellow ('New question this lesson raised for me'). On the green note, they write	Distribute sticky notes. Say: 'You have 3 minutes — one green note, one yellow note. Be specific on your green note: name the physics process, not just the topic.' WAIT 3 minutes in silence. Then invite students to post notes. Read 4 green notes aloud — for each, ask the class: 'Does this claim connect the physics to the phenomenon? Give me a thumbs up if you think it does.' After clustering yellow questions, point to the cluster	Driving Question Board as a Living Knowledge Artefact: The DQB makes collective understanding visible and cumulative. Green notes document growing explanatory power; yellow notes surface authentic student curiosity that drives the next lesson's inquiry. Clustering yellow questions in front of students shows that confusion is productive and that science is an ongoing process — directly modelling	Teacher reads all green sticky notes after class. Observable indicator: Does the note name a specific physics mechanism (IR absorption, re-emission, back-radiation, enhanced greenhouse effect) rather than a vague statement ('gases cause warming')? Notes with only vague statements identify students who need a brief re-engagement at the start of Lesson 4. Yellow notes are catalogued to inform lesson

📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	one specific claim they can now support with evidence from today's simulation (e.g., 'More CO_2 in the air above Kenya absorbs more infrared radiation and sends it back to the surface, raising temperatures — this is why the Lewis Glacier is melting'). On the yellow note, they write one new question (e.g., 'If CO_2 has always been in the air, why did glaciers only start melting recently? How much CO_2 is too much?'). Students post both notes on the DQB. The teacher reads 4–5 green notes aloud and clusters the yellow questions by theme, previewing: 'These questions about WHEN the change accelerated and HOW MUCH warming is already locked in — that is exactly where we are going in Lesson 4.'	about 'how much CO_2 is too much' and say: 'This cluster is asking about FEEDBACK and TIPPING POINTS. Write those two words in your notebook. We will come back to them.' Photograph the DQB board with the school tablet/phone to preserve the state before the next lesson.	NGSS Science & Engineering Practice 1: Asking Questions.	planning for Lessons 4–5.
Model Building Phase 📹 **VIDEO:** [Video: Molecules and Light] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students open their cumulative 'Atmospheric Model' page — a running diagram they have been building since Lesson 1. In Lesson 1 they drew the physical layers of the atmosphere. In Lesson 2 they added arrows showing sources of greenhouse gas emissions from Kenyan activities. Today, in Lesson 3, they add the energy-flow layer: (a) yellow arrows for incoming shortwave solar radiation passing through the atmosphere to the surface; (b) red upward arrows for IR radiation emitted by the surface; (c) orange curved arrows from GHG molecules showing re-emission in all directions, including back toward the surface; (d) a thermometer symbol at the surface labelled 'Enhanced Greenhouse Effect → $\Delta T > 0$'; (e) a dotted box around the Lewis Glacier at high altitude with the annotation:	Say: 'Open your Atmospheric Model page. You are a climate physicist updating a model as new evidence comes in — just as scientists at the Kenya Meteorological Department do when new satellite data arrives.' Project a completed teacher exemplar on the board showing the three arrow types. Say: 'Your model must have all three arrow types, all five vocabulary labels, and the Lewis Glacier annotation. You have 8 minutes.' CIRCULATE: stop at 4–5 students and ask: 'Point to the place in your model where the ENHANCED greenhouse effect is different from the natural greenhouse effect — what has changed?' WAIT 10 seconds for each student to point and explain before moving on. With 2 minutes remaining, say: 'Complete your sentence stem at	Cumulative Model Revision (Iterative Modelling — NGSS Science & Engineering Practice 2): Adding a new layer to an existing model rather than starting fresh each lesson mirrors how scientific models are genuinely built — incrementally, with each new piece of evidence refining rather than replacing prior understanding. The physical act of annotating the same diagram across lessons builds a concrete, student-owned record of growing explanatory power. The sentence stem forces metacognitive reflection: students must consciously articulate how the physics model connects to the real-world phenomenon.	Teacher reviews collected model pages after the lesson. Three-level rubric — (1) Emerging: at least two arrow types drawn but unlabelled or incorrectly connected to GHG molecules; (2) Developing: all three arrow types drawn and labelled, sentence stem present but vague; (3) Proficient: all three arrow types, all five vocabulary terms correctly used, sentence stem draws a specific causal connection between IR back-radiation and Lewis Glacier melt or rainfall disruption. Students at Emerging level receive a brief targeted re-engagement card at the start of Lesson 4.

	'Increased ΔT drives glacial melt — Lewis Glacier −92% since 1934.' Students annotate each arrow with the correct physics term. At the bottom of the model page, they complete the sentence stem: 'This model helps explain the anchoring phenomenon because ___.' The teacher collects model pages at the end of the lesson for review before Lesson 4.	the bottom. Make it specific — connect the physics arrows to the phenomenon.' Collect all model pages as an exit artefact.		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) MECHANISM vs DESCRIPTION — When students wrote their 'Explain It to a Farmer' paragraphs, did the majority include a true causal chain (human activity → GHG → IR absorption → back-radiation → temperature rise → glacier melt), or did they describe the greenhouse effect without explaining the physics mechanism? If most students described rather than explained, plan a 10-minute re-engagement at the start of Lesson 4 using the 'Photon Absorption' tab of the PhET simulation with two targeted questions: 'Why does CO_2 absorb IR but N_2 does not?' and 'What happens to the absorbed energy?'. (2) MISCONCEPTION PERSISTENCE — Did the ozone-layer/greenhouse-effect conflation persist after Phase 3? Check the DQB green notes and model pages. If more than 30% of students still conflate the two, create a T-chart comparison card for Lesson 4. (3) DEVICE/CONNECTIVITY ISSUES — If the PhET simulation was run as a teacher-led projector demonstration rather than individual/pair investigation, did students have sufficient opportunity to manipulate variables themselves? If not, consider assigning the simulation as a home investigation task (via ARES offline platform) before Lesson 4, framing it as: 'Be a scientist: find the CO_2 concentration that raises the temperature by exactly 2°C above the pre-industrial baseline.' (4) KENYA CONNECTION DEPTH — Did students draw explicit connections between specific Kenyan GHG sources (identified in Lesson 2) and the IR absorption mechanism in this lesson? If the connection felt abstract, open Lesson 4 with a 5-minute 'link-back' activity: display the Lesson 2 list of Kenyan GHG sources and ask students to annotate each source with which greenhouse gas it produces and what infrared effect that gas has.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed in the PhET Greenhouse Effect simulation that: (a) removing the atmosphere causes surface temperature to drop dramatically; (b) increasing CO_2 and CH_4 concentrations raises equilibrium surface temperature; (c) CO_2 and CH_4 molecules absorb infrared photons and re-emit them in all directions, while N_2 molecules do not absorb IR photons at all; (d) shortwave (visible) radiation passes through greenhouse gas molecules without being absorbed, but longwave infrared radiation emitted by the surface is absorbed by GHG molecules.
What did I learn?	Students learned the physics mechanism of the greenhouse effect: incoming shortwave solar radiation passes through the atmosphere and is absorbed by Earth's surface; the surface re-emits this energy as longwave infrared radiation; greenhouse gas molecules (CO_2, H_2O vapour, CH_4) absorb this IR radiation and re-emit it in all directions, including back toward the surface — this 'back-radiation' raises surface temperature above what it would be with no atmosphere (the natural greenhouse effect); human activities in Kenya (charcoal burning, deforestation of the Mau, cattle ranching) have increased GHG concentrations, enhancing back-radiation beyond natural levels (the enhanced greenhouse effect), which explains the rising temperatures driving Mount Kenya's Lewis Glacier retreat.
How does this explain the phenomenon?	Students can now explain: WHY Mount Kenya's Lewis Glacier is melting (enhanced back-radiation from elevated CO_2 and CH_4 concentrations raises atmospheric and surface temperatures above the glacier's equilibrium freezing point, driving ice loss); WHY

	the Embu County farmer's rainfall is becoming unpredictable (higher surface temperatures increase evaporation rates from Lake Victoria and the Indian Ocean, altering the moisture and energy dynamics of the long-rains system, disrupting the March–May precipitation patterns that Kenyan farmers have relied on for generations). Students can trace a complete causal chain from a specific Kenyan human activity to an observable climate impact on the landscape and farming community.

LESSON 4 (40 minutes): How Is Global Warming Changing Kenya's Environment?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will analyse observational and data-based evidence to identify and explain the cascading environmental and social effects of global warming in Kenya, connecting the physics mechanism learned in prior lessons to real local consequences.
Knowledge	Students can explain how enhanced greenhouse gas concentrations lead to rising average surface temperatures; identify at least four observable effects of global warming in Kenya (Lewis Glacier retreat, Rift Valley drought, Kisumu flooding, disrupted tea harvests in Kericho, unpredictable long rains); and describe how a change in one Earth system drives changes in other systems.
Skills	Students can read and interpret NASA global temperature anomaly graphs and sea-level rise charts; construct an effects web that maps causal chains from the greenhouse effect mechanism to local Kenyan consequences; and use scientific vocabulary to revisit and reinterpret a primary source testimony.
Attitudes	Students demonstrate curiosity and concern about the real-world consequences of atmospheric physics for Kenyan communities, and show empathy toward farmers, pastoralists, and others already experiencing climate impacts.
Key Inquiry Question	How does the greenhouse effect influence global and local climates?
Purpose in Storyline	This lesson is the evidence-gathering pivot of the unit. Students now have the physics mechanism (Lessons 1 and 2), the emission sources (Lesson 2), and the ozone context (Lesson 3). Lesson 4 asks: what has the enhanced greenhouse effect actually done to Kenya so far? By building an effects web and returning to the Embu farmer testimony with new scientific language, students develop the causal chain that will be fully synthesised in Lesson 7 and that directly answers the driving question.
Safety Notes	No laboratory hazards. If discussing drought, flood, or food insecurity, be sensitive to students who may have personally experienced these events in their own families or counties. Allow students to engage at their own emotional comfort level.

B. LESSON OVERVIEW
Students open by returning to the Embu farmer testimony from Lesson 1 and making a fresh prediction about which specific physical changes are responsible for the disrupted rains. They then observe NASA global temperature anomaly data and a short National Geographic video clip showing glacier retreat, sea-level rise, and extreme weather events. Using a Lake Victoria supporting phenomenon — rising lake levels that flooded Kisumu homes in 2020 — students link global warming to a local event they likely know. In the Explain phase, student groups construct an effects web on large paper, tracing causal chains from enhanced back-radiation to surface warming to glacier melt, altered rainfall, drought, and flooding across Kenya. The lesson closes with students returning to the Embu farmer testimony and rewriting one sentence of it in precise scientific language, then adding new sticky notes to the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Graph labels and	Students are shown the Embu farmer testimony from Lesson 1 printed on a card on each desk.	Teacher reads the testimony slowly and then says: We have spent three lessons learning about	Activation of prior learning through a concrete human testimony as anchor. Students must translate	Teacher scans notebook entries during pair discussion and listens for whether students are

[scales](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Ocean Currents (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	The teacher reads it aloud slowly: the long rains now arrive weeks later, maize and bean yields have fallen, and the farmer says the seasons their grandparents relied on no longer match what happens today. Students write in their notebooks: List two or three specific physical changes in the atmosphere or environment that you now think — based on Lessons 1 to 3 — could be causing what this farmer is describing. Students then share predictions with a partner before three or four pairs share aloud. Predictions are recorded on the whiteboard in a column labelled Our Predictions.	how our atmosphere traps energy and how human activities are changing it. Now I want you to act like a scientist who is asked to explain this farmers experience. What specific physics have we learned that could be at work here? Give students 90 seconds of silent writing time before pair discussion. After pairs share, teacher annotates the whiteboard predictions without evaluating them, saying: We will test these predictions against real data in a few minutes. Keep them in mind.	abstract physics concepts into an explanation for a real observed consequence, surfacing their current mental models before new evidence is introduced.	connecting the enhanced greenhouse effect and rising temperatures to rainfall disruption — or whether they are still describing only the ozone layer or direct sunlight effects. Misconceptions about ozone depletion causing warming are noted for targeted correction in the Explain phase.
Observe Phase 📹 **VIDEO:** [Graph labels and scales](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Ocean Currents (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Students view two data displays in sequence. First, the teacher projects the NASA GISS Surface Temperature Analysis global temperature anomaly graph (1880 to present), pre-downloaded and printed as a handout if internet is unreliable. Students are asked: What pattern do you see from 1950 onward? Mark the steepest rise on your copy. Second, the teacher shows a two-minute clip from a National Geographic video on glacier retreat and sea-level rise, pausing at two moments: when a Kenyan highland region is referenced and when flooding of low-lying areas is shown. After the clip, teacher introduces the Lake Victoria supporting phenomenon: between 2019 and 2020, Lake Victoria rose by approximately 1.2 metres above its long-term average, flooding thousands of homes in Kisumu and displacing fishing communities. Students are given a simple cause-chain card with three boxes and an arrow	While students examine the NASA graph, teacher circulates and asks: Can you find the year when the anomaly first exceeded 0.5 degrees Celsius? What was happening in Kenya around that time in terms of industrialisation or population growth? After the video clip, teacher says: The Lake Victoria flooding is not a random disaster. Scientists have linked it to changes in rainfall patterns over the Lake Victoria basin driven by warmer regional temperatures. Your job now is to trace the chain of physics that connects a CO_2 molecule absorbing infrared radiation to a family in Kisumu losing their home. Allow 2 minutes of silent cause-chain writing before moving on.	Data-to-phenomenon bridging: students move from global quantitative data (temperature anomaly graph) to a vivid visual phenomenon (glacier retreat and flooding video) to a specific local supporting phenomenon (Lake Victoria rise and Kisumu floods). The cause-chain card structures the cognitive move from observation to causal reasoning without yet demanding full explanation.	Teacher collects three or four cause-chain cards and reads them aloud anonymously, asking the class: Does this chain make physical sense at each step? What is missing? This surfaces gaps in causal reasoning (for example, students who skip the step of rising surface temperature and jump directly from CO_2 to flooding) that the Explain phase will address.

	between each, labelled Step 1, Step 2, Step 3, and asked: Use what you just saw to fill in one causal chain that starts with more greenhouse gases in the atmosphere and ends with a Kenyan community being affected.			
Explain Phase 📹 **VIDEO:** Graph labels and scales Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Ocean Currents (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Student groups of four receive a large sheet of paper and a set of labelled sticky notes in four colours: orange for causes, yellow for physical processes, green for environmental effects, and pink for social or economic effects on Kenyan communities. Groups construct an effects web starting from the central node Enhanced Greenhouse Effect and branching outward. They must include at least one chain ending at Mount Kenya Lewis Glacier retreat, one ending at disrupted long rains in Embu County, one ending at Rift Valley drought, one ending at Kisumu or Lake Victoria flooding, and one ending at Kericho tea harvest disruption. Students use their prior lesson notes, the NASA handout, and the cause-chain cards. After 10 minutes of group construction, each group does a gallery walk — moving to one other groups web and adding one sticky note of a connection they think is missing, using a different colour pen.	Teacher circulates during web construction and uses targeted questions: How does a rise in average surface temperature change evaporation rates over the Indian Ocean? How does that change what happens to the clouds that bring the long rains to Embu? If students are only listing effects without showing physics links, teacher prompts: Can you draw an arrow and write the physical process that connects these two nodes? After the gallery walk, teacher brings the class together and says: I noticed several groups connected warming temperatures to both more flooding in one place and more drought in another. How can warming cause both of these at the same time? Allow 30 seconds of wait time before taking responses. Teacher guides students to the explanation that warming intensifies the water cycle — increasing evaporation and making rainfall more intense but also more spatially uneven — so some areas flood while others dry out.	Effects web construction externalises student mental models and makes causal chains visible and discussable. The gallery walk introduces peer critique and adds cross-group connections, building a richer collective model. The teacher-facilitated discussion of the apparent paradox of simultaneous flooding and drought deepens mechanistic understanding and corrects oversimplified linear thinking.	Teacher photographs each groups effects web for later review. During the whole-class discussion, teacher listens for whether students can articulate the intensification of the water cycle as the mechanistic link between warming and both drought and flood. Students who cannot are identified for additional scaffolding in Lesson 5.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Graph labels and scales Source: Khan	Students return to the Driving Question Board displayed at the front of the room. The teacher reminds them that in Lesson 1 they posted questions and initial predictions. Each student now receives two sticky notes. On the first, they write one question from	Teacher says: Look at the questions you and your classmates posted in Lesson 1. Some of you asked why the rains are getting unpredictable. Some asked whether the glacier melt and the rain changes are connected. Can you answer those now? Write	The DQB update makes the students growing understanding visible and metacognitive. By requiring scientific vocabulary in the answered-question notes, the teacher formalises language development. New questions sustain inquiry momentum and	Teacher reviews all DQB sticky notes after class to assess the quality of scientific language and the coherence of causal explanations. Students who are still writing only surface-level answers (for example, the glacier is melting because it is hot) without

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Ocean Currents (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Lesson 1 that they can now partially or fully answer, and write a two-sentence answer using scientific vocabulary from Lessons 1 through 4. On the second, they write one new question that todays lesson has raised for them — something they still do not understand or want to investigate. Students post both notes in the appropriate columns on the DQB: Answered (or Partly Answered) and New Questions. The teacher reads four or five of the answered questions aloud and briefly affirms the scientific language used, then highlights two or three new questions that preview Lesson 5 on mitigation and adaptation.	your answer in the language of a physicist — use words like infrared radiation, surface temperature anomaly, water cycle intensification, and cascading effects. After students post notes, teacher reads one answered note aloud and says: This student used the phrase intensified water cycle to explain both flooding and drought. That is precisely the kind of scientific reasoning we are building toward in this unit. For new questions, teacher says: Several of you are asking what we can actually do about this. That is exactly what Lesson 5 will address.	signal to students that scientific learning is cumulative and open-ended.	mechanistic language are targeted for a brief check-in at the start of Lesson 5.
Model Building Phase 📹 **VIDEO:** Graph labels and scales Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Ocean Currents (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student opens their cumulative model notebook page — started in Lesson 1 and revised in Lessons 2 and 3. They now add a new layer to the right side of their model diagram labelled Observed Effects in Kenya. Using their effects web as a guide, they draw at least four arrows leading from the central process box (Enhanced Back-Radiation raises surface temperature) to four labelled effect boxes: Lewis Glacier retreat, disrupted long rains in Embu, Lake Victoria rise and Kisumu flooding, and Kericho tea harvest disruption. On each arrow, they write the physical or environmental process that connects cause to effect (for example: higher surface temperature increases evaporation over Indian Ocean, intensifying moisture supply but altering timing of monsoon winds reaching inland Kenya). Finally, students return to the Embu farmer testimony card and underline one sentence. They	Teacher says: Your model has grown across four lessons. In Lesson 1 you drew the sun and some arrows. Now you should be able to trace energy from the sun all the way to a farmer in Embu who cannot predict when to plant their maize. Make sure every arrow on your model has a label — a label that names the physical process, not just says leads to. Circulate and prompt students who are drawing effects without labelling the connecting process. For the testimony rewrite, teacher says: If the farmer says the rains arrive later now, what would a physicist say is happening to the atmospheric circulation patterns that deliver those rains? Encourage students to use the phrase water cycle intensification and inter-tropical convergence zone shift if they encountered it in the video or class discussion. Collect two or three notebooks at the end of class to review the testimony rewrites as formative	Cumulative model revision consolidates learning across all four lessons into a single evolving artefact. The requirement to label every arrow with a physical process prevents superficial listing of effects and demands mechanistic thinking. The testimony rewrite task bridges scientific and everyday language, reinforcing that physics explains real human experiences.	Teacher reviews collected notebooks for two criteria: (1) are the four effect boxes present and connected with labelled process arrows? and (2) does the testimony rewrite use at least two pieces of precise scientific vocabulary correctly? Feedback is returned at the start of Lesson 5. Students who have correctly traced the causal chain from enhanced back-radiation through surface temperature rise to water cycle intensification to disrupted rainfall are assessed as having met the lesson SLO.

	rewrite that sentence in their notebook using precise scientific language, replacing everyday words with physics vocabulary.	evidence.		

D. TEACHER REFLECTION
After the lesson, consider: Did students successfully distinguish between the greenhouse effect mechanism (physics of IR absorption and re-emission) and its downstream effects (glacier retreat, flooding, drought, harvest disruption)? Were students able to explain how warming can simultaneously cause flooding in Kisumu and drought in the Rift Valley — the intensification paradox — or did most students still think of warming as a single uniform change? Did the effects web gallery walk generate genuine scientific dialogue between groups, or did students simply copy each others nodes without critique? Were the DQB answered-question notes written in scientific language that reflects Lessons 1 through 4, or are students still relying on pre-instructional vocabulary? Which students are struggling to make the leap from global temperature data to local Kenyan consequences, and how will Lesson 5 provide additional scaffolding for them? Note any students whose testimony rewrite demonstrated exceptional causal reasoning for recognition in Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that NASA global surface temperature data shows a sharply rising anomaly from 1950 to the present, and that the Lake Victoria water level rose approximately 1.2 metres between 2019 and 2020, flooding homes in Kisumu. We also observed that the Lewis Glacier on Mount Kenya has retreated dramatically since 1934, and that Kericho tea harvests and Embu long rains have become increasingly unpredictable.
What did I learn?	We learned that a rise in average global surface temperature — caused by enhanced back-radiation from increased greenhouse gas concentrations — drives cascading effects across multiple Earth systems: it accelerates glacier melt, intensifies the water cycle (increasing evaporation and altering monsoon timing), causes some regions to experience more flooding while others experience more drought, and disrupts the agricultural seasons that Kenyan farmers have relied on for generations.
How does this explain the phenomenon?	We explained that the Embu farmer is experiencing disrupted long rains because rising surface temperatures over the Indian Ocean alter evaporation rates and shift the inter-tropical convergence zone, changing when and where moisture-laden winds reach inland Kenya. The same warming mechanism that is melting the Lewis Glacier is intensifying and destabilising the water cycle that governs Kenyan rainfall — so both phenomena share a single physical cause: the enhanced greenhouse effect driven by rising greenhouse gas concentrations.

LESSON 5 (2 lessons (80 minutes)): How Do Human Activities Amplify the Greenhouse Effect — and What Does This Mean for Mount Kenya and Kenyan Farmers?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students gather and analyse evidence linking human-driven increases in greenhouse gas concentrations to enhanced warming, glacier retreat on Mount Kenya, and disrupted rainfall patterns across Kenyan farming regions.
Knowledge	By the end of this lesson, learners should be able to: (a) explain how human activities — burning fossil fuels, deforestation, and industrial processes — increase concentrations of greenhouse gases (CO_2, CH_4, N_2O, water vapour); (b) describe how increased GHG concentrations amplify the greenhouse effect and raise average global and regional temperatures; (c) connect enhanced warming to observed consequences including glacial retreat on Mount Kenya and shifting precipitation patterns affecting maize and bean farmers in Embu, Meru, and Nyeri counties.
Skills	Analysing graphs of atmospheric CO_2 concentration (Keeling Curve) and temperature anomaly data; comparing timelines of human industrial activity with observed climate data; constructing evidence-based written and diagrammatic explanations; evaluating the strength of evidence linking cause to effect.
Attitudes	Appreciate that physics and chemistry together explain large-scale environmental changes; demonstrate responsibility by acknowledging the role of human choices in driving climate change; show respect for the lived experiences of Kenyan farmers whose livelihoods are affected.
Key Inquiry Question	How do human activities increase the greenhouse effect, and how does this explain the melting of Mount Kenya's glaciers and the unpredictability of Kenyan farmers' rains?
Purpose in Storyline	Lessons 1–4 established what electromagnetic radiation is, how the greenhouse effect works naturally, how the ozone layer interacts with UV radiation, and how energy balance governs Earth's temperature. Lesson 5 is the pivotal evidence-gathering lesson in which students shift from 'how does the natural greenhouse effect work?' to 'what has changed, why has it changed, and who is responsible?' Students examine real CO_2 concentration data, global temperature anomaly records, and human emissions timelines, then directly link these to the Mount Kenya glacier photographs and the Embu farmer's testimony introduced in Lesson 1. This lesson is the evidential turning point of the entire sequence.
Safety Notes	No hazardous materials are used in this lesson. If internet-connected devices are used to access PhET or LabXchange simulations, ensure students access only approved URLs. Remind learners that discussions about climate responsibility should remain respectful and evidence-based; the topic can be emotionally sensitive for students whose families are farmers directly affected by climate variability.

B. LESSON OVERVIEW

Students open by revisiting their Driving Question Board predictions from Lesson 1 and reviewing the glacier paired images and Embu farmer testimony. They then complete a structured data-analysis activity using the Keeling Curve (atmospheric CO_2 1958–present) and a global temperature anomaly graph, identifying correlations with industrial milestones. A PhET or LabXchange simulation ('Greenhouse Effect') allows students to manipulate GHG concentrations and observe resulting temperature changes. Students then apply their findings to construct a written causal chain — from charcoal burning and vehicle emissions in Nairobi to amplified greenhouse warming to glacial melt on Mount Kenya and shifting long rains in Embu County. The lesson closes with a Driving Question Board update and a cumulative model revision adding the 'human amplification' layer to their atmospheric energy model.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Video: Molecules and Light] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students revisit the anchoring phenomenon: the paired Lewis Glacier images (1934 vs. present) and the Embu maize farmer's testimony displayed on the classroom wall or screen. Each student silently re-reads their own Lesson 1 prediction card (kept in their science notebooks) answering: 'What is causing these changes?' Students then write a 3-sentence updated prediction responding to the prompt: 'Based on everything you have learned in Lessons 1–4 about radiation, the greenhouse effect, energy balance, and the ozone layer — what do you now think is the most likely cause of glacier retreat on Mount Kenya and the disruption of rains in Embu? What evidence would convince you?' Students share predictions with a shoulder partner (Think-Pair-Share) before three are cold-called to share with the class.	Display the paired glacier images and read aloud the Embu farmer's testimony. Say: 'We have been building our understanding for four lessons. Before we look at new evidence today, I want you to look back at your very first prediction from Lesson 1. How has your thinking changed — and what are you still unsure about?' Allow 4 minutes of silent individual writing. Then say: 'Turn to your shoulder partner and compare your updated predictions. What do you agree on? Where do you still disagree?' Allow 2 minutes of pair talk. Apply 12-second WAIT TIME after posing the cold-call prompt before selecting students. Cold-call exactly 3 students — aim for one student who is confident, one who is uncertain, and one whose Lesson 1 prediction was markedly different from scientific consensus. Record key prediction phrases on the board under the heading 'Our Updated Predictions — Lesson 5 Start.' Say: 'Keep these in mind — by the end of today you will know whether the evidence supports them.'	Prediction Revision (Metacognitive Anchoring): Students explicitly compare their current understanding to their Lesson 1 naïve conception. This surfaces conceptual growth, identifies persistent misconceptions (e.g., 'the ozone hole is causing warming'), and creates cognitive readiness for the new evidence. Connecting back to the phenomenon at the start of every lesson reinforces the storyline thread and gives each new piece of evidence narrative purpose.	Collect updated prediction cards at the end of the phase. Observable indicators of readiness: (a) students reference specific mechanisms from Lessons 1–4 (e.g., 'IR radiation is trapped by greenhouse gases'), not just vague statements like 'pollution'; (b) students identify at least one thing they are still unsure about, indicating productive epistemic humility; (c) cold-call responses reveal whether students conflate ozone depletion with the greenhouse effect — a common misconception to address explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** [Video: Molecules and Light] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:**	PART A — Graph Analysis (30 minutes): In groups of three, students receive a printed or on-screen data pack containing: (1) the Keeling Curve showing atmospheric CO_2 concentration from 1958 to the present (labelled in ppm), annotated with key human milestones (post-WWII industrialisation, rapid motorisation of Nairobi 1970s–1990s, rise of Chinese manufacturing 2000s,	Distribute data packs and worksheets before the lesson to save time. As groups work on Part A, circulate and use these targeted questions: 'What does ppm mean? How many CO_2 molecules are we talking about in every million molecules of air?' (Do NOT answer immediately — allow 10 seconds of group think first.) 'Where on this graph does the rate of increase appear to get steeper? What was	Data Literacy with Contextual Anchoring: Students practice reading and interpreting real scientific datasets (Keeling Curve, temperature anomaly) rather than simplified textbook charts. Annotating the graphs with Kenyan and global human milestones connects abstract data to historical and local context. The simulation provides a controlled manipulative experience that isolates the causal	Collect completed worksheets. Observable indicators: (a) students correctly describe both upward trends and note their temporal correlation; (b) students articulate the difference between correlation and causation unprompted or when probed; (c) simulation observation sentences use the causal language 'because' and reference IR radiation trapping, not just 'more gases = warmer'; (d)

Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Kenya's expanding charcoal trade); (2) NASA/NOAA global average temperature anomaly data from 1880 to present; (3) a simplified timeline of global fossil fuel extraction volumes from 1800 to present. Students complete a structured analysis worksheet: — 'Describe the trend in CO_2 concentration from 1958 to present. What was the concentration in 1958? What is it approximately today?' — 'Describe the trend in global average temperature anomaly from 1900 to present. When did the anomaly become consistently positive?' — 'Mark on both graphs the period 1950–2000. What happened to both variables during this period? What was happening in human society during the same period?' — 'What pattern do you notice between CO_2 concentration and temperature anomaly? Does correlation prove causation? What else would you need to know?' PART B — PhET / LabXchange Simulation (15 minutes): Each student (or pair if devices are limited) opens the PhET 'Greenhouse Effect' simulation. Students: — Set atmosphere to 'Today's atmosphere' and record the equilibrium temperature. — Add 'Lots of greenhouse gases' and record the new temperature. — Remove all greenhouse gases and record the temperature. — Record observations in their notebooks using the sentence frame: 'When I increased greenhouse gas concentration, the temperature _______ because _______.' PART C — Kenyan Emissions Context (5 minutes): Students read a short paragraph (provided on the worksheet)	happening in Kenya and the world at that time?' Cold-call one group to share their correlation observation and immediately ask a second group: 'Do you agree that correlation proves causation? What would a scientist say?' For Part B, demonstrate the PhET simulation on the projector for 90 seconds before releasing students to work independently. Say: 'You are now going to do what atmospheric physicists do — change one variable and observe the effect. Record everything — even if it surprises you.' Circulate during simulation time; if a student claims 'the temperature barely changed,' prompt: 'Try the extreme setting — what happens then? Why might the real world change more slowly than the simulation?' For Part C, read the Kenya emissions paragraph aloud to the whole class and pause after: 'Kenya contributes very little to global emissions — yet Kenyan farmers and Kenyan glaciers are paying the price.' Ask: 'Is that fair? Write one sentence about what this makes you think or feel.' Allow 12 seconds of WAIT TIME before taking 2 unprompted responses.	variable (GHG concentration) from confounding factors — directly addressing the correlation-vs-causation question raised in the worksheet. The Kenya emissions paragraph introduces the concept of climate justice, deepening engagement and emotional investment in the phenomenon.	students who write something substantive (even if emotionally charged) in response to the Kenya fairness prompt demonstrate engagement with the human dimensions of the phenomenon. Flag any group that still conflates ozone depletion with the greenhouse effect for direct intervention during the Explain phase.

	explaining that Kenya's CO_2 emissions per capita are low compared to industrialised nations, but that Kenya is acutely vulnerable to the effects of global emissions — the Lewis Glacier has lost over 90% of its mass since 1900, and the Kenya Meteorological Department has documented a 2–3 week delay in the onset of the long rains since the 1980s.			
Explain Phase 📹 **VIDEO:** [Video: Molecules and Light] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students individually construct a written causal chain in their notebooks using the following scaffold provided on the board: 'Step 1 → Human activities (name at least TWO specific ones relevant to Kenya) release _______ into the atmosphere. Step 2 → Increased concentration of _______ gases means that more _______ radiation emitted by Earth's surface is _______. Step 3 → Less energy escapes to space, so Earth's energy balance _______. Step 4 → Global average temperature _______. Step 5 → On Mount Kenya, this means _______. Step 6 → For a maize farmer in Embu County, this means _______.' After 8 minutes of individual writing, students share their causal chains in groups of four, identifying: (a) steps where all four members agree; (b) steps where members used different language — and deciding which language is most scientifically precise; (c) any step that no one in the group feels confident about. Groups then nominate one step they want the teacher to clarify. The class reconvenes; the teacher addresses the top 2–3 nominated unclear steps using the analogy of a blanket thickening around the	After observe phase, say: 'You now have three sources of evidence: two graphs, a simulation, and a real Kenyan data point about the glacier and the rains. Your job now is to connect all of that evidence into one coherent explanation — a causal chain from a human action to a physical outcome on Mount Kenya and in Embu.' Display the 6-step scaffold on the board and read each step aloud. Say: 'Be specific — don't just write 'pollution.' Name the gas. Name the process. Name the radiation type.' Allow 8 minutes of silent individual writing. Circulate and quietly prompt any student who is stuck on Step 1: 'What do matatus in Nairobi burn? What do farmers burn when they clear land? What do charcoal jikos release?' After group sharing, conduct a whole-class gallery of Step 6 responses by asking 3 cold-called students to read their farmer-in-Embu sentence. After each, say: 'What physics word or concept did they use? Did anyone use a different piece of physics to explain the same outcome?' To address the ozone misconception, say explicitly: 'I want to address something I have seen in some of your writing. Some of you wrote	Causal Chain Scaffold with Peer Calibration: The 6-step scaffold forces students to make their reasoning explicit and sequential, preventing them from skipping the physical mechanism (IR absorption) and jumping straight from 'burning fuel' to 'glacier melts.' Group comparison of causal chains develops scientific language precision and peer accountability. The explicit misconception address uses the contrast strategy — placing the correct and incorrect explanations side by side so students can clearly distinguish them. The exam-style question provides authentic assessment practice while embedding the phenomenon as the answer context.	Collect individual causal chains and exam-style answers. Observable indicators of mastery: (a) Step 2 correctly names CO_2, CH_4, or N_2O and specifies IR (infrared) radiation, not UV or visible light; (b) Step 3 references energy balance or the term 'net energy gain'; (c) Steps 5 and 6 are specific and connected (e.g., 'increased temperatures accelerate glacial melt, reducing dry-season water flow to rivers that irrigate farms in Meru and Embu' — not just 'the glacier melts and farmers suffer'); (d) the exam-style answer contains at least 3 of: named gas, IR trapping, energy balance disruption, temperature rise, physical consequence on glacier. Students whose chains are missing the radiation mechanism should be flagged for small-group follow-up.

	Earth. Students also explicitly address and correct the common misconception that 'the ozone hole is causing warming' by distinguishing: ozone blocks UV (Lesson 3) while greenhouse gases trap IR (Lessons 2 and 4). Students finish by writing one exam-style short answer: 'Explain, using physics, why burning charcoal in Nairobi contributes to the melting of glaciers on Mount Kenya. (4 marks)'	that the ozone hole is causing warming. Let us be precise: what does the ozone layer do? [WAIT 10 seconds — cold-call 2 students.] And what do greenhouse gases do? [WAIT 10 seconds — cold-call 2 different students.] These are two different mechanisms. Both are real. Both matter. But they are not the same thing.' For the exam-style question, allow 5 minutes, then cold-call one student to read their answer aloud and ask the class: 'What would you add? What mark out of 4 would you give this answer and why?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives two sticky notes of different colours. On the FIRST (green) sticky note, they write one specific piece of evidence from today's lesson that helps answer the driving question: 'Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable?' On the SECOND (orange) sticky note, they write one new question that today's evidence has raised for them — something they still do not understand or want to know more about. Students post both sticky notes on the class Driving Question Board under the column 'Lesson 5 — Human Amplification.' The teacher reads 5–6 of the new questions aloud to the class and sorts them into two clusters on the board: 'Questions about causes' and 'Questions about responses/solutions.' The teacher previews: 'Your questions about what we can do about this — those are exactly what Lessons 6 and 7 will address. You are thinking like climate scientists.'	Hand out sticky notes before the DQB phase begins. Say: 'The green note is your evidence contribution — what did you learn today that counts as real evidence for answering our driving question? Make it specific: mention a number, a graph, a gas, or a physical process. Vague answers like 'CO_2 causes warming' without detail will not earn full credit.' Allow 3 minutes of individual writing. Say: 'The orange note is your intellectual honesty note. What are you genuinely still unsure about? What new question did today's lesson raise for you? There are no wrong questions here — the best scientists always have more questions than answers.' Allow 2 minutes. As students post, spend 90 seconds reading the orange notes and grouping them. Say: 'I am seeing two kinds of questions emerging. [Hold up the causes cluster.] Some of you are still asking HOW — how exactly does CO_2 trap heat, how long does it stay in the atmosphere. [Hold up the	DQB Evidence Posting with Question Sorting: The two-colour sticky note protocol separates evidence consolidation (what do I now know?) from inquiry extension (what do I still want to know?), preventing students from conflating comprehension with understanding. Reading and sorting new questions aloud models the scientific practice of question refinement and shows students that unanswered questions are the engine of further inquiry, not a sign of failure. Previewing Lessons 6–7 maintains narrative momentum in the storyline.	Observable indicators: (a) green notes contain specific, physics-grounded evidence (e.g., 'CO_2 has risen from 315 ppm in 1958 to over 420 ppm today, and temperature anomaly has risen by about 1.1°C in the same period'); (b) orange notes reveal the frontier of each student's understanding — collect and photograph the DQB before the next lesson to inform lesson 6 planning; (c) the distribution of 'causes' vs. 'responses' questions shows whether the class is ready to move to solutions (Lessons 6–7) or needs more consolidation of mechanisms.

		responses cluster.] Others of you are asking WHAT NOW — what can Kenya do, what can individuals do, is it already too late? Both kinds of questions are scientifically and humanly important. Let's note which lessons will address each.' Circle the 'responses' cluster and write 'Lessons 6 & 7' next to it.		
Model Building Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative atmospheric energy model drawn in their notebooks across Lessons 1–4. The model currently shows: the sun emitting shortwave radiation; Earth absorbing and re-emitting IR; greenhouse gases absorbing and re-emitting IR; the ozone layer filtering UV. Students now add a new layer to the model — the 'Human Amplification Layer' — by: (1) drawing upward arrows from the Earth's surface labelled 'CO_2, CH_4, N_2O — increased by human activity'; (2) thickening or darkening the greenhouse gas layer to represent increased concentration; (3) drawing a new arrow showing 'increased IR re-emitted back to Earth's surface'; (4) adding a label at the surface: 'Net energy gain → rising surface temperature'; (5) adding two consequence boxes at the bottom of the model connected by arrows to 'rising surface temperature': Box A — 'Mount Kenya: Lewis Glacier retreats as ice-melt rate exceeds accumulation rate'; Box B — 'Embu County: disrupted Hadley cell circulation → delayed and unpredictable long rains.' Students annotate the model with at least three specific numbers drawn from today's evidence (e.g., CO_2 concentration values, temperature anomaly magnitude, percentage of	Say: 'Your model has been growing for five lessons. Today we add the most important layer yet — the human fingerprint. Open your notebooks to your cumulative model. You have five minutes to add the Human Amplification Layer. Use your causal chain from the Explain phase to guide you. Every arrow must be labelled. Every consequence box must name a specific Kenyan location.' Allow 5 minutes of individual model revision. Circulate and check for: arrows drawn in the correct direction (back toward Earth's surface, not away from it); the consequence boxes referencing the phenomenon by name (Mount Kenya glacier, Embu farmer); numerical annotations. After 5 minutes, say: 'Now swap your model with the person next to you. Use the three-point checklist on the board to assess their model. Write one specific piece of feedback — not 'good job' — something they could add or clarify.' Allow 3 minutes of peer assessment. Cold-call two students to share the feedback they received and how they will revise their model as a result. Close by saying: 'Your model now shows how physics connects a matatu's exhaust pipe in Nairobi, to a layer of gas in the	Cumulative Model Revision with Peer Assessment: The running model across all seven lessons is the primary sense-making artefact of the unit. Each lesson's revision makes abstract mechanisms visible and permanent in the learner's notebook. Adding specific numbers forces students to move from qualitative to quantitative reasoning. Peer assessment using a structured checklist develops evaluative skills and distributes the formative assessment load. The teacher's closing statement explicitly names the cross-scale explanatory power of physics, reinforcing the purpose of the entire sequence.	Collect or photograph student models before Lesson 6. Observable indicators of mastery: (a) arrows correctly show increased IR re-emission back toward Earth's surface — not toward space; (b) the human amplification layer is distinct from the natural greenhouse effect layer drawn in earlier lessons; (c) both Kenyan consequence boxes are present and reference the phenomenon by name; (d) at least three specific numerical values are annotated (e.g., '315 ppm CO_2 in 1958,' '+1.1°C global anomaly by 2023,' '90% of Lewis Glacier area lost since 1900'); (e) peer feedback is specific and constructive, not generic. Models lacking the directionality of the IR re-emission arrow should be flagged for Lesson 6 warm-up correction.

	glacier area lost). Groups briefly share their completed models and peer-assess using a 3-point checklist: (1) Is the human amplification mechanism shown? (2) Are the two Kenyan consequences both present? (3) Are there at least three specific numbers?	atmosphere, to a melting glacier on Mount Kenya, to an unpredictable rain season for a farmer in Embu. That is the power of physics — it explains things across scale, across distance, and across time.'		

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions: (1) Did students successfully distinguish between the ozone layer mechanism (UV absorption, Lesson 3) and the greenhouse effect mechanism (IR trapping, Lessons 2 and 5)? If the misconception persisted in causal chains or exam-style answers, plan a 5-minute explicit contrast activity at the start of Lesson 6. (2) Were students able to read the Keeling Curve and temperature anomaly graphs independently, or did they need more scaffolding? If graph literacy was a barrier, consider adding a worked example of reading a two-axis graph to the Lesson 6 warm-up. (3) How did students respond emotionally and intellectually to the Kenya climate justice paragraph — the observation that Kenya contributes little to global emissions but bears significant consequences? Were responses thoughtful and evidence-grounded, or did the discussion become unproductive? Use this to gauge readiness for the solutions-focused Lessons 6 and 7. (4) Review the orange DQB sticky notes: which student questions are most urgent? Do students need more time on mechanisms, or are they ready to move to responses and mitigation? Adjust Lesson 6 pacing accordingly. (5) Were simulation devices available and functioning? If PhET was inaccessible, note which groups completed the simulation section and which did not — plan a brief simulation catch-up or whole-class demonstration at the start of Lesson 6 to ensure all students have the manipulative experience.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students analysed the Keeling Curve (CO_2 concentration 1958–present) and a global temperature anomaly graph, identifying a strong upward correlation between rising atmospheric CO_2 and rising global average temperatures. Students also used the PhET Greenhouse Effect simulation to observe that increasing greenhouse gas concentrations raises equilibrium surface temperature. Students noted that Kenya's Lewis Glacier has lost over 90% of its mass since 1900, and that the Kenya Meteorological Department has recorded a 2–3 week delay in the onset of the long rains since the 1980s.
What did I learn?	Human activities — including burning fossil fuels (petrol, diesel, charcoal), deforestation, and industrial processes — release additional CO_2, CH_4, and N_2O into the atmosphere, increasing greenhouse gas concentrations above natural levels. Higher GHG concentrations mean more infrared radiation emitted by Earth's surface is absorbed and re-emitted back toward the surface, reducing the amount of energy escaping to space. This creates a net energy gain at Earth's surface, raising average global temperatures. This enhanced greenhouse effect is the physical mechanism behind the retreat of Mount Kenya's Lewis Glacier and the disruption of the long rains experienced by farmers in Embu, Meru, and Nyeri counties.
How does this explain the phenomenon?	The observed rise in both atmospheric CO_2 concentration and global average temperature since industrialisation is explained by the physics of infrared absorption: greenhouse gas molecules (CO_2, CH_4, N_2O) absorb outgoing IR radiation that would otherwise escape to space and re-emit it in all directions, including back toward Earth's surface. When human activities increase the concentration of these gases, this absorption is enhanced — the atmospheric 'blanket' thickens — and more energy is retained in the Earth system. The resulting increase in surface temperature drives glacial melt on Mount Kenya by raising ablation rates above accumulation rates, and disrupts regional precipitation patterns by altering atmospheric circulation and the timing of the Inter-

	Tropical Convergence Zone (ITCZ), which governs Kenya's long and short rain seasons.

LESSON 6 (80 minutes): The Ozone Layer: A Separate Shield — and How Its Damage Connects to Mount Kenya's Disappearing Ice	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the structure and function of the ozone layer, distinguish ozone depletion from the greenhouse effect, and evaluate how both atmospheric processes interact to drive the changes observed at Mount Kenya and on Kenyan farms.
Knowledge	Students explain that the ozone layer is a region of the stratosphere (15–35 km altitude) that absorbs harmful UV-B radiation; that chlorofluorocarbons (CFCs) and other halocarbons catalytically destroy ozone molecules; that ozone depletion and the greenhouse effect are distinct but interacting atmospheric phenomena; and that increased UV-B levels affect ecosystems, human health, and surface energy balance.
Skills	Students analyse real UV-B irradiance data from the UNEP Ozone Data Portal for Nairobi; compare and contrast ozone depletion with the greenhouse effect using a structured T-chart; evaluate the effectiveness of the Montreal Protocol as a policy response; and revise their cumulative atmospheric model to incorporate the ozone layer as a separate but related shield.
Attitudes	Students appreciate the importance of international scientific cooperation (Montreal Protocol) in reversing atmospheric damage; demonstrate intellectual humility by recognising that the atmosphere is more complex than previously modelled; and show environmental responsibility by connecting ozone recovery to local Kenyan futures.
Key Inquiry Question	How does the greenhouse effect influence global and local climates? What are the causes and effects of climate change, and what measures can be taken to address it?
Purpose in Storyline	In Lessons 1–5, students built evidence that greenhouse gases trap outgoing infrared radiation, warming Earth's surface — explaining the glacier retreat and disrupted rains. In Lesson 6, students discover that the atmosphere has a second protective function — the ozone layer absorbs incoming UV-B — and that this layer is separately threatened by CFCs. By adding the ozone layer to their model, students gain a more accurate and complete picture of 'what the atmosphere does,' and recognise that atmospheric protection is multi-layered, fragile, and responsive to human chemistry — deepening their answer to the driving question.
Safety Notes	No hazardous chemicals or apparatus are used in this lesson. Students work with digital resources and printed data sheets. Remind students that UV-B radiation is harmful to skin and eyes, and that this is a real risk in Nairobi due to its equatorial position and elevation — encourage students to take sun-safety habits seriously. Ensure all devices are charged and internet access (or pre-downloaded resources) is confirmed before the lesson begins.

B. LESSON OVERVIEW

In this 80-minute lesson — the sixth in a seven-lesson evidence-gathering sequence — Grade 10 Physics students investigate the ozone layer as a distinct but interacting atmospheric shield. Beginning with a provocative re-examination of the anchoring phenomenon (Why are UV levels in Nairobi rising even as we try to reduce warming?), students first predict what the ozone layer does and what might damage it. They then gather evidence by working through a structured Khan Academy lesson on ozone chemistry and interrogating real UV-B irradiance data from the UNEP Ozone Data Portal for Nairobi. Using a T-chart sensemaking tool, students distinguish ozone depletion from the greenhouse effect — noting that the former involves UV absorption in the stratosphere while the latter involves IR trapping in the troposphere, but that both are worsened by industrial chemicals. Students evaluate the Montreal Protocol as a successful policy case, contrasting it with the slower progress on greenhouse gases, and revise their cumulative model to show the ozone layer as a separate stratospheric shield. The Driving Question Board is updated to capture this new layer of complexity. The lesson prepares students for Lesson 7's synthesis and final model presentation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Ozone layer and eukaryotes show up in the Proterozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown two pieces of evidence side by side on the board or screen: (1) a graph of UV-B irradiance measured in Nairobi between 1990 and 2020 showing a peak in the late 1990s followed by a gradual decline, and (2) a world map showing the Antarctic ozone hole at its maximum extent. Students are asked to respond individually in their science notebooks to three prompt questions: 'What do you think the ozone layer does for Earth? What do you predict caused the UV-B spike in Nairobi? Do you think ozone depletion and the greenhouse effect are the same thing, different things, or related things — explain your reasoning.' Students then share predictions with a shoulder partner for 2 minutes before three cold-called pairs share with the class. All predictions are recorded on a 'Prediction Anchor Chart' at the front of the room — to be revisited at the end of the lesson.	Display the UV-B graph and ozone hole map simultaneously. Say: 'Before we read or watch anything, I want to know what YOU think. Take 4 minutes — write your predictions in your notebook. No wrong answers right now.' Start a 4-minute timer. After 4 minutes: 'Turn to your shoulder partner. You have 2 minutes — compare your predictions.' After 2 minutes, cold-call exactly 3 pairs using the class register (select students who have not been cold-called recently). For each pair, say: 'What did you two agree on — and where did you disagree?' After each response, allow 10 seconds of wait time before commenting. Annotate the Prediction Anchor Chart with student language — use their exact words, not corrected scientific terminology yet. Close with: 'Hold those predictions — by the end of this lesson, you'll be able to tell me which ones were right, which ones need adjusting, and why.'	Prediction Anchor Chart — students externalise prior knowledge and misconceptions before instruction. Recording exact student language (not corrected terms) creates a visible baseline that can be returned to, making conceptual change explicit and metacognitively visible. This activates prior knowledge from Lessons 1–5 (greenhouse effect) and surfaces the common misconception that ozone depletion and the greenhouse effect are the same phenomenon.	Circulate during individual writing (2 minutes). Look for: (1) students who correctly identify that the ozone layer absorbs UV radiation — note these students for targeted questioning in Phase 3; (2) students who conflate ozone depletion with the greenhouse effect — note these students for explicit address during the T-chart activity; (3) students who leave notebook prompts blank — prompt them with: 'What do you remember about the atmosphere from our last five lessons? Start there.' Collect 4–5 notebooks at random to review prediction entries after the lesson.
Observe Phase 📹 **VIDEO:** Ozone layer and eukaryotes show up in the Proterozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Students gather evidence from two sources in sequence. SOURCE A (25 minutes): Working in pairs, students work through a structured excerpt of the Khan Academy lesson 'Ozone Layer and Ozone Hole' (pre-loaded on devices or printed as a guided reading sheet if internet is unavailable). As they work, they complete a structured Evidence Collection Table with four columns: 'What the ozone layer is made of,' 'What it does	Before Source A, say: 'You have 25 minutes for Source A. Your job is NOT to copy — your job is to understand well enough to explain it to someone who wasn't here. Write in your own words. Go.' Set a 25-minute timer. Circulate every 4–5 minutes, pausing at pairs who are copying verbatim and redirecting: 'Close the screen/text for a moment — what did that section mean in your own words?' At the 20-minute mark, give a 5-	Structured Evidence Collection Table — a two-source scaffolded note-taking strategy that separates evidence gathering from interpretation. By using a four-column table, students organise new information into causal chains (what ozone is → what it does → what damages it → what was done) before being asked to compare or contrast. The graph annotation task then connects abstract chemistry to real Nairobi	During Source A circulation, check that the 'What damages it' column reflects the catalytic cycle of CFCs (ozone destroyed, CFC molecule regenerated) — a key conceptual point. Students who write only 'CFCs destroy ozone' without noting the catalytic/cyclical nature need a prompt: 'Read that section again — does the CFC molecule get used up, or does it keep working? Why does that matter?' During Source B annotation, look

Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(what radiation does it block and where),' 'What damages it and how (the chemistry of CFC catalysis in simple terms),' and 'What has been done about it.' Students write in their own words — not copy-pasted. SOURCE B (15 minutes): Students then access the UNEP Ozone Data Portal (pre-loaded dataset for Nairobi, or teacher-printed data sheet) and read a simplified UV-B irradiance trend graph for Kenya/East Africa. Students annotate the graph: 'When did UV-B peak? When did the Montreal Protocol take effect? What does the trend after 2000 suggest?' Students record three observations from the data in their Evidence Collection Table.	minute warning: 'Five minutes — make sure all four columns have something in them.' Transition to Source B: 'Now we're looking at real data from Nairobi. Scientists at UNEP have been measuring UV-B levels in East Africa for decades. Your job: read the graph and annotate it — three observations minimum. What story does this data tell?' Allow 15 minutes. At the midpoint (7–8 minutes in), cold-call one student: 'Wanjiru, what year does the data show the UV-B peak? And what was happening politically around that time that might explain the change after that?' Allow 10 seconds of wait time. If the student is unsure, say: 'Check your Source A notes — what did the Montreal Protocol say and when?' Do not give the answer directly.	UV-B data, grounding the lesson in the local phenomenon and making the evidence consequential. This mirrors scientific practice: scientists cross-reference experimental data with known mechanisms.	for students who can connect the post-2000 decline in UV-B to the Montreal Protocol's effect — this is the key causal inference. Students who cannot make this link need scaffolding: 'What year did the Montreal Protocol begin reducing CFC production? Now look at the graph — what happens to UV-B after that lag period?'
Explain Phase 📹 **VIDEO:** Ozone layer and eukaryotes show up in the Proterozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students complete a structured T-chart in their notebooks titled: 'Ozone Depletion vs. Greenhouse Effect — Same or Different?' The T-chart has eight comparison rows: (1) Which layer of the atmosphere is involved, (2) What type of radiation is affected, (3) What gases are responsible, (4) What human activities cause it, (5) What the effect on Earth's surface temperature is, (6) What the effect on ecosystems/health is, (7) What international agreements address it, (8) What the current trend is (getting better or worse?). Students complete the T-chart individually (8 minutes), then discuss in groups of four to reconcile differences (5 minutes). Each group then identifies ONE way the two phenomena interact — how does ozone depletion affect or relate to climate change?	Distribute the T-chart (or write the eight rows on the board). Say: 'You have 8 minutes to fill this in on your own — use your Evidence Collection Table from Phase 2 and your notes from Lessons 1–5. This is a thinking task, not a copying task.' Set an 8-minute timer. After 8 minutes: 'Now, in your group of four, compare your T-charts. Where do you disagree? Spend 5 minutes reconciling.' After group discussion, say: 'Each group — I want ONE sentence that describes how these two phenomena interact. Not just what each one does separately — how do they affect each other?' Cold-call four groups in sequence. After each response, wait 10 seconds, then ask the class: 'Does the next group agree? What would you add or change?' After all four groups have shared, synthesise on the board:	Structured T-chart with Venn synthesis — the T-chart forces precise comparison across eight parallel dimensions, preventing the most common misconception (that ozone depletion and greenhouse effect are the same). The subsequent Venn diagram synthesis elevates students from comparison to integration, requiring them to identify causal connections between the two systems. This models the scientific practice of constructing explanations from multiple evidence sources. The final pivot back to the anchoring phenomenon ensures that new knowledge is applied to the specific Kenyan context rather than remaining abstract.	Collect T-charts from two or three groups to review after class. During whole-class discussion, listen for: (1) students who correctly distinguish stratosphere (ozone) from troposphere (greenhouse effect) — these students demonstrate spatial understanding of the atmosphere; (2) students who can name CFCs as both ozone-depleting substances AND greenhouse gases — this is the key interaction insight; (3) students who can articulate that the greenhouse effect (not ozone depletion) is the primary driver of glacier retreat and rainfall disruption at Mount Kenya — this demonstrates conceptual precision. If fewer than 60% of cold-called students can make this distinction, pause and re-teach using the cumulative model on the board before

	Groups share interactions with the class. Teacher facilitates a whole-class discussion that arrives at the key insight: while distinct, ozone depletion and the greenhouse effect interact (e.g., CFCs are also potent greenhouse gases; a cooler stratosphere due to greenhouse warming can worsen ozone depletion; UV-B damage to phytoplankton reduces a key carbon sink). Teacher then explicitly connects back to the anchoring phenomenon: 'So — does ozone depletion explain Mount Kenya's glaciers melting? Partially? Fully? Let's be precise.'	draw a two-circle Venn diagram with 'Ozone Depletion' and 'Greenhouse Effect' and populate the overlap with student-generated interactions. Then pivot: 'Let's go back to our phenomenon. Mwangi the farmer in Embu — his rain is unpredictable. Amina at the base of Mount Kenya — the glacier is shrinking. Which of these two phenomena — greenhouse effect, ozone depletion, or both — best explains what they're experiencing? Be specific. Write one sentence.' Cold-call 3 individual students. Wait 10 seconds after each response before commenting.		proceeding to Phase 4.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Ozone layer and eukaryotes show up in the Proterozoic eon] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board, which has been building across Lessons 1–5. Each student receives two sticky notes of different colours — one colour for 'new evidence we now have' and one colour for 'new questions this lesson raised.' Students individually write their contributions (3 minutes), then place them on the DQB under the appropriate columns: 'Evidence for the Driving Question' and 'New Questions.' The teacher reads aloud selected sticky notes and facilitates a 5-minute whole-class discussion that tracks the arc of the inquiry across Lessons 1–6. Students are prompted to identify which parts of the driving question they can now answer with confidence, which parts remain uncertain, and what Lesson 7 will need to address. The class also notes on the DQB that today's lesson revealed the atmosphere is more complex than their earlier model — it has multiple protective functions that are separately	Hand out two sticky notes per student. Say: 'You have 3 minutes — on your first sticky note, write one piece of new evidence from today that helps answer our driving question about Mount Kenya's glaciers and the farmers' rains. On your second sticky note, write one new question that today's lesson opened up for you — something you're still wondering about.' Set a 3-minute timer. After students place their notes, read aloud four or five — select at least one that captures the ozone-greenhouse interaction, one that reflects confusion, and one that points forward to Lesson 7. Say: 'Look at our DQB — in Lesson 1, we asked whether the same cause could explain both the glacier and the rain. We now have six lessons of evidence. What can we say confidently? What do we still need?' Cold-call three students. Then say: 'In Lesson 7, our final lesson, you will build your complete model and prepare your explanation. What do you still need	Two-colour sticky note DQB update — using two distinct colours for 'evidence' and 'new questions' maintains the epistemic distinction between what is known and what is not, which is central to scientific inquiry. The cumulative DQB makes the growth of understanding visible across the sequence, turning the board into a public record of the class's shared knowledge construction. Reading selected notes aloud validates student thinking and models how scientists communicate findings incrementally.	Scan all sticky notes before the class discussion. Look for: (1) evidence notes that correctly describe the ozone layer's function and distinguish it from the greenhouse effect — these show successful lesson learning; (2) evidence notes that still conflate the two phenomena — these students need targeted follow-up in Lesson 7; (3) question notes that point toward legitimate scientific extensions (e.g., 'If ozone is recovering, will UV-B in Nairobi keep falling?' or 'Does a warmer atmosphere always mean more ozone depletion?') — affirm these publicly as 'expert questions.' Photograph the updated DQB for the teacher's records.

	vulnerable.	to make that model complete?' Accept student responses without correction — use them to preview Lesson 7.		
Model Building Phase 📹 **VIDEO:** [Ozone layer and eukaryotes show up in the Proterozoic eon] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative atmospheric model from their science notebooks — the diagram they have been updating since Lesson 1, which currently shows: incoming solar radiation, the troposphere with greenhouse gases, Earth's surface, and outgoing IR radiation being partially trapped. Today, students add two new features to their model: (1) A stratospheric ozone layer (labelled, with altitude approximately 15–35 km) shown absorbing incoming UV-B radiation before it reaches the surface — drawn as a distinct coloured band above the troposphere; (2) Annotations showing CFCs rising from Earth's surface, reaching the stratosphere, and catalytically destroying ozone molecules — with a note that fewer ozone molecules = more UV-B reaching the surface. Students also add a colour-coded key distinguishing UV-B (absorbed by ozone), visible light (passes through to surface), and infrared (trapped by greenhouse gases). Students write a two-sentence model caption: 'My model now shows that the atmosphere has [X] because [Y]. This helps explain the Mount Kenya phenomenon because [Z].' Students share captions with their shoulder partner and make any final revisions. Selected models are photographed by the teacher for the portfolio record.	Say: 'Take out your atmospheric model — the one you've been building since Lesson 1. Today it gets a major upgrade.' Give students 10 minutes to add the ozone layer and CFC annotations. Circulate and ask targeted questions: 'Where exactly in your diagram is the ozone layer — is it above or below the greenhouse gas layer? Good — why does that matter?' and 'Show me on your diagram what UV-B does if there's plenty of ozone. Now show me what happens if CFCs have thinned the ozone layer.' After 10 minutes: 'Write your two-sentence caption now — 2 minutes.' After captions are written, say: 'Read your caption to your shoulder partner. Does their caption agree with yours about what the atmosphere does? Where are they different?' Allow 2 minutes for partner discussion. Cold-call two students to read their captions aloud. For each, ask the class: 'Thumbs up if you agree with that caption — thumbs sideways if you'd change something.' Address one or two suggested changes publicly, modelling how scientists revise models based on peer critique. Close with: 'Your model now shows two separate but related atmospheric shields. In Lesson 7, you'll use this complete model to build your final explanation for our driving question. Protect that notebook.'	Cumulative model revision — the practice of returning to and revising a single running model across multiple lessons embeds the scientific practice of developing and revising models based on new evidence (NGSS Science and Engineering Practice 2). The colour-coded radiation key forces students to differentiate UV-B, visible light, and IR — the three radiation types central to this unit — in a single spatial diagram. The two-sentence caption transforms the visual model into a verbal explanation, requiring students to articulate mechanism, not just structure. Peer sharing of captions creates a low-stakes peer-review process that mirrors scientific practice.	As you circulate during model building, look for three specific features: (1) the ozone layer is drawn in the stratosphere, clearly above the tropospheric greenhouse gas layer — students who draw it at the surface or conflate the two layers need immediate redirection; (2) the model distinguishes UV-B (blocked by ozone) from IR (trapped by greenhouse gases) — students who draw all radiation the same need prompting: 'These are different types of radiation — how do you show that difference in your diagram?'; (3) the two-sentence caption correctly identifies the ozone layer as a separate shield and connects its thinning to increased UV-B at the surface (not to warming). Collect 5–6 model notebooks to photograph before the next lesson and use them to plan Lesson 7 targeted support.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) CONCEPTUAL CLARITY — Were students able to distinguish ozone depletion from the greenhouse effect by the end of Phase 3? If fewer than 70% of students made this distinction clearly in their T-charts, plan a 5-minute re-teach at the start of Lesson 7 using the Venn diagram before moving to synthesis. (2) MODEL QUALITY — Do the revised cumulative models correctly show the ozone layer in the stratosphere, above the greenhouse gas layer in the troposphere? If models are spatially confused, use a projected labelled diagram at the start of Lesson 7 to anchor spatial understanding before the final model is constructed. (3) DQB PROGRESSION — Do the sticky notes suggest that students understand the driving question is becoming answerable — that they now have sufficient evidence about both the greenhouse effect (Lessons 1–5) and the ozone layer (Lesson 6) to build a complete explanation? If DQB questions are still very foundational (e.g., 'What is the greenhouse effect?'), revisit those students' understanding before Lesson 7's synthesis. (4) KENYA RELEVANCE — Did students connect the UV-B data from Nairobi to their own lived experience (sun exposure, skin health, agricultural impacts)? If not, consider opening Lesson 7 with a brief local news item about UV index warnings in Nairobi to re-ground the inquiry. (5) POLICY REASONING — Did students engage meaningfully with the Montreal Protocol as a success story? The contrast between the Montreal Protocol (ozone recovering) and slower progress on greenhouse gas reduction (climate still worsening) is a powerful lesson in science-policy interaction — note whether students spontaneously made this comparison, as it will be important for Lesson 7's final discussion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A UV-B irradiance trend graph for Nairobi (1990–2020) showing a peak in the late 1990s and gradual decline post-2000; a world map of the Antarctic ozone hole at maximum extent; and a structured Khan Academy lesson excerpt on ozone chemistry including the catalytic destruction of ozone by CFCs.
What did I learn?	The ozone layer is a stratospheric shield (15–35 km altitude) that absorbs incoming UV-B radiation before it reaches Earth's surface; it is chemically distinct from the tropospheric greenhouse gas layer that traps outgoing infrared radiation; CFCs (chlorofluorocarbons) catalytically destroy ozone molecules — one CFC molecule can destroy thousands of ozone molecules; the Montreal Protocol (1987) has successfully reduced CFC production and the ozone layer is gradually recovering; CFCs are also potent greenhouse gases, meaning ozone depletion and the greenhouse effect are distinct but interacting phenomena; and rising UV-B levels in Nairobi during the 1990s are real, measurable evidence of ozone depletion's local impact in Kenya.
How does this explain the phenomenon?	The atmosphere performs at least two separate but related protective functions: (1) the ozone layer in the stratosphere absorbs harmful UV-B radiation, and (2) greenhouse gases in the troposphere regulate surface temperature by trapping infrared radiation. Both functions are disrupted by industrial chemicals — CFCs damage ozone directly and also act as greenhouse gases. The primary cause of Mount Kenya's glacier retreat and the disruption of Kenyan farmers' rainfall patterns is the enhanced greenhouse effect (driven by CO_2, CH_4, and other greenhouse gases), not ozone depletion — but ozone depletion adds a second layer of atmospheric vulnerability that affects ecosystems, human health, and indirectly interacts with climate. The Montreal Protocol demonstrates that when the world acts on clear scientific evidence, atmospheric healing is possible — raising the question of whether similar international action on greenhouse gases can produce comparable results.

LESSON 7 (40 minutes): Final Model Building, Driving Question Board Closure, and Unit Synthesis: Answering the Mystery of Mount Kenya's Retreating Glaciers	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students synthesise all prior learning across Lessons 1 to 6 into a single coherent cumulative model that explains the full causal chain from solar energy input through atmospheric greenhouse trapping, to global warming, to local Kenyan consequences, to mitigation pathways — and use that model to write a complete scientific explanation of the anchoring phenomenon.
Knowledge	Students will demonstrate knowledge of: (1) the mechanism of the greenhouse effect — solar shortwave radiation passes through the atmosphere, Earth re-emits longwave infrared radiation, and greenhouse gases such as CO2, CH4, and N2O absorb and re-radiate that infrared energy, warming the lower atmosphere; (2) human activities in Kenya that increase greenhouse gas concentrations, including charcoal burning, deforestation of Mau Forest, and livestock emissions; (3) the distinct but interacting role of the ozone layer in absorbing UV-B radiation and how CFC-driven ozone depletion compounds atmospheric damage; (4) observed effects of global warming on Kenya, including Lewis Glacier retreat, disrupted long rains in Embu, Lake Victoria level changes, and Turkana droughts; and (5) mitigation and adaptation strategies at national and household scale in Kenya.
Skills	Students will: construct and refine a multi-layered annotated scientific model connecting atmospheric physics to local Kenyan climate impacts; synthesise evidence from across six lessons into a coherent written scientific explanation; compare models across peer groups to identify consensus explanations and remaining uncertainties; and evaluate which components of their Driving Question Board have been answered and which remain open.
Attitudes	Students will demonstrate scientific curiosity by engaging seriously with remaining uncertainties; show responsibility by connecting their learning to personal and community-level climate action; and display appreciation for the value of cumulative scientific inquiry in addressing real-world problems facing Kenyan communities.
Key Inquiry Question	How does the greenhouse effect influence global and local climates, and what measures can be taken to address climate change — as evidenced by the retreat of Mount Kenya's glaciers and the disruption of rainfall patterns experienced by Embu farmers?
Purpose in Storyline	This lesson is the capstone of the seven-lesson unit. Students return to the exact images and testimony that opened Lesson 1 — the 1934 photograph of the Lewis Glacier alongside the current satellite image, and the Embu farmer testimony about unpredictable long rains — and this time they write a full, evidence-based scientific explanation using every layer of their cumulative model. The Driving Question Board is formally closed: questions posed in Lesson 1 are matched to the explanations developed across the unit. This lesson transforms the unit from a collection of separate lessons into a coherent scientific story, fulfilling the promise made to students in Lesson 1 that physics could explain something deeply important about their own country.
Safety Notes	No laboratory hazards in this lesson. Ensure that class discussion remains emotionally supportive when students discuss climate impacts on farming communities, as some students may have personal family connections to subsistence farming or pastoral communities affected by climate change. Treat the Embu farmer testimony and Turkana pastoralist testimonies from Lesson 6 with respect and seriousness.

B. LESSON OVERVIEW

In this 40-minute capstone lesson, students revisit the anchoring phenomenon — the retreat of Mount Kenya's Lewis Glacier and the Embu farmer's testimony about unpredictable long rains — and use their cumulative scientific model to write a full explanation of both changes. The lesson moves through five structured phases: a Predict phase in which students individually recall and sketch the key causal chain before peer discussion; an Observe phase in which the original Lesson 1 images and

testimony are displayed again alongside a visual summary timeline of evidence gathered across all six lessons; an Explain phase in which students complete their final annotated cumulative model and write their scientific explanation; a Driving Question Board closure phase in which the class matches answers to original questions and surfaces new questions that remain open; and a Model Building phase in which groups present their final models, identify consensus, and celebrate the journey of inquiry. The lesson honours the full CBE competency profile: knowledge application, critical thinking, communication, collaboration, and citizenship.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students sit individually for 4 minutes with a blank A4 sheet. The teacher displays the original anchoring phenomenon images on the board — the 1934 Lewis Glacier photograph and the current satellite image — and reads aloud the Embu farmer testimony from Lesson 1 verbatim: 'The long rains used to come in March, like clockwork. My grandmother planted by the sound of the first thunder. Now I wait until April, sometimes May, and still I am not sure. The maize does not know what to do.' Students are asked to sketch, from memory, the full causal chain that connects human activities to glacier retreat and rainfall disruption. They should not look at their notebooks. This activates retrieval practice and reveals what has been durably learned. Students then turn to a partner for 2 minutes and compare their chains, noting any steps their partner included that they had forgotten.	Before displaying the images, the teacher says: 'I am going to take you back to the very first day of this unit. Do you remember what you saw? Do you remember what you did not understand? In a moment I will show you those same two images and read you those same words from the farmer in Embu. This time, I want you to sketch — without any notes — the full scientific story that explains what is happening. Use arrows, labels, anything you like. You have four minutes, working alone.' Display the images. Read the testimony slowly and clearly. Then set a visible timer for 4 minutes. Circulate silently — do not prompt or hint. After 4 minutes, say: 'Now turn to your partner. Share your chains. Look for any step your partner included that surprised you or that you had missed. You have two minutes.' After partner sharing, ask 2 to 3 volunteers: 'What was the most important step that your partner reminded you of?' WAIT TIME: After each volunteer responds, wait 5 seconds before commenting, allowing other students to react first.	Retrieval practice sketch activates prior learning across all six lessons as a connected causal chain rather than isolated facts. Partner comparison surfaces gaps and reinforces the most critical links in the chain — particularly the connection between infrared absorption by greenhouse gases (Lesson 1), human emission sources (Lesson 2), ozone layer interaction (Lesson 3), observed warming effects (Lesson 4), and mitigation pathways (Lessons 5 and 6). The teacher circulates to note which links students omit most frequently, which informs targeted emphasis in the Explain phase.	Teacher observes sketches during circulation to identify: (1) whether students include the infrared re-radiation mechanism or only name CO_2 as a cause; (2) whether students connect both glacier retreat and rainfall disruption to the same root cause; (3) whether students include mitigation and adaptation steps or treat the model as purely descriptive. Students who include only one or two causal links will receive additional scaffolding in the Explain phase through the provided model template.
Observe Phase 📹 **VIDEO:** Video: Molecules	The teacher displays a structured 'Evidence Timeline' on the board — a visual strip showing the six lessons as six panels, each with its	Display the Evidence Timeline prominently. Say: 'This strip tells the story of what we have investigated over seven lessons.	The Evidence Timeline externalises the full unit narrative as a visible object, helping students see the coherence of	Teacher listens for: (1) whether groups can articulate the infrared mechanism in their own words when answering question 1; (2)

[and Light](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	key evidence item: Lesson 1 — PhET simulation showing infrared absorption; Lesson 2 — Kenyan charcoal burning and Mau Forest deforestation data; Lesson 3 — ozone layer UV-B shield and CFC data from Nairobi; Lesson 4 — NASA temperature data and Lewis Glacier retreat measurements; Lesson 5 — Olkaria Geothermal and Lake Turkana Wind Farm mitigation data; Lesson 6 — Turkana pastoralist and Mombasa coastal community testimonies. Students spend 5 minutes in groups of four reviewing the timeline and answering three focus questions written on the board: (1) Which piece of evidence from across all six lessons do you find most convincing that human activity is warming Earth? (2) Which link in the causal chain — from greenhouse gas emissions to glacier retreat — do you think is hardest to explain to someone who has never studied physics? (3) What evidence connects the physics of the atmosphere to the life of the Embu farmer specifically?	Each panel is one piece of evidence we gathered together. In your groups, discuss the three questions on the board. You have five minutes. I want every person in the group to speak at least once.' Circulate and listen. When you hear a group struggling with question 3 — the link between atmospheric physics and the Embu farmer — pose a probing question without answering it: 'What happens to the water cycle when surface temperatures rise? Where does Embu get its rainfall from? Think about what we learned about evaporation and shifting rainfall patterns in Lesson 4.' WAIT TIME: After posing this probe, step back and allow 8 seconds before any further support. After 5 minutes, ask one group to share their answer to question 1 and a different group to share their answer to question 3. Acknowledge both with: 'That is exactly the kind of scientific reasoning this unit has been building toward.'	their learning journey. The three focus questions are deliberately tiered: question 1 is evaluative (which evidence is most convincing), question 2 is communicative (which link is hardest to explain), and question 3 is integrative (connecting physics to lived local experience). This mirrors the competency structure of CBE and prepares students for the scientific explanation writing in the Explain phase.	whether groups identify the water cycle disruption or shifting inter-tropical convergence zone as the mechanism connecting atmospheric warming to Embu rainfall — if not, this becomes a targeted input in the Explain phase; (3) whether any groups reference the ozone layer distinction correctly, showing they have retained the nuance from Lesson 3.
Explain Phase 📹 **VIDEO:** [Video: Molecules and Light](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Each student receives a structured Final Explanation Sheet with two components. Component A is a labelled cumulative model template: an outline diagram showing the sun, Earth's atmosphere (divided into troposphere and stratosphere), Earth's surface, and arrows for incoming solar radiation and outgoing infrared radiation. Students must annotate the diagram with: the names and sources of key greenhouse gases relevant to Kenya; the mechanism of infrared absorption and re-	Distribute the Final Explanation Sheet. Say: 'This is your moment to show everything you have learned. Component A is your model — annotate it fully. Component B is your explanation — write it as if you are explaining to a scientist from another country who has never been to Kenya but wants to understand what is happening here. Be precise. Be Kenyan. Use real places and real data where you can.' Set a timer for 10 minutes. Circulate actively. For students who are stuck on the mechanism, use the prompt: 'Start	The structured Final Explanation Sheet scaffolds but does not over-scaffold: the diagram template provides spatial organisation while requiring students to generate all content from memory and reasoning. The written explanation prompt requires integration across multiple lessons and demands use of local Kenyan specificity, which prevents students from writing generic global warming descriptions copied from textbooks. Peer feedback using sentence starters trains scientific communication as a skill and	Teacher collects Component B written explanations at the end of the lesson as the primary summative evidence of learning. During circulation, the teacher uses a simple three-level rubric mentally: Level 1 — student names greenhouse gases and glacier retreat but does not explain the infrared mechanism; Level 2 — student explains the mechanism and connects it to at least one Kenyan cause and one Kenyan effect; Level 3 — student explains the full causal chain including water cycle disruption for rainfall,

	radiation; the ozone layer position and UV-B absorption function; the connection between enhanced greenhouse effect and surface temperature rise; and at least two observed effects on Kenya (glacier retreat and rainfall disruption). Component B is a written scientific explanation of 150 to 200 words responding to the prompt: 'Using the physics of the atmosphere, explain why Mount Kenya's Lewis Glacier has retreated since 1934 and why the long rains in Embu County have become harder to predict. Your explanation must include the greenhouse effect mechanism, at least one Kenyan human activity that contributes to the problem, and at least one action that Kenyans are taking in response.' Students work individually for 10 minutes, then share their written explanation with a partner for peer feedback using two sentence starters: 'I can see the physics clearly when you write...' and 'One part that could be more precise is...'	with the sun. What kind of radiation does it send? What does Earth do with that energy? What do greenhouse gases do with it?' For students who omit mitigation, say: 'Your explanation says what is happening and why — but what are Kenyans doing about it? Your model needs that layer too.' After individual writing, say: 'Share your Component B with your partner. Use the two sentence starters on the sheet to give feedback. Be honest and be kind — this is scientific peer review.' WAIT TIME: During peer feedback, resist the urge to correct. Observe and note strong explanations to share with the class afterward.	surfaces misconceptions that the teacher can address in the DQB phase.	references specific Kenyan mitigation actions, and uses precise physical language (infrared radiation, re-radiation, energy balance). This rubric informs differentiated feedback on returned scripts.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES	The original Driving Question Board from Lesson 1 is displayed — either the physical board if it has been maintained in the classroom, or a photograph of it projected on the screen. Students can see the questions, predictions, and wonderings they posted in Lesson 1 using sticky notes or cards. In groups, students receive three coloured cards: green (We can now answer this question fully), yellow (We can partly answer this question but some uncertainty remains), and red (This is a new or deeper question that our learning has raised). Groups spend 5 minutes categorising the	Display the original DQB. Say: 'In Lesson 1, you asked questions you could not yet answer. You posted predictions you were not sure about. Now you are scientists who have spent seven lessons investigating. Let us see what you now know.' Distribute the coloured cards. Say: 'Green means you can fully answer it with evidence. Yellow means you have a partial answer but honest uncertainty remains. Red means a brand new question your learning has opened up — because good science always opens new doors.' After posting, facilitate the gallery walk. Then ask: 'Which question on this	The DQB closure is a metacognitive ritual that externalises the growth of understanding over the unit. The three-colour categorisation system requires students to make honest evaluations of what they know and what remains uncertain — a core scientific practice. The red card requirement ensures the lesson does not close with false completeness: science always generates new questions, and this is modelled explicitly. The gallery walk creates a shared class view of collective knowledge and remaining uncertainty, reinforcing that science is a community	Teacher notes: (1) how accurately groups categorise questions — misclassifying a yellow as green reveals overconfidence and can be addressed with a gentle challenge: 'What evidence would you need to be fully certain about that?'; (2) the quality and sophistication of red cards — superficial red cards (for example, 'Will the glacier melt completely?') suggest surface-level engagement, while deep red cards (for example, 'If methane is 80 times more potent than CO2 over 20 years, why do climate policies focus more on CO2?') suggest genuine scientific thinking; (3) whether students can articulate

for similar readings	original DQB questions and writing at least one red card — a new question that has emerged from their deeper understanding. Groups then post their colour-coded cards next to the original questions. The class does a brief gallery walk (2 minutes) to read other groups' categorisations and red cards.	board would you say is the most important one we answered? And which red card — which new question — excites you most?' Take 2 to 3 responses. For the most powerful new question raised by a student, say: 'That question — [repeat it] — is exactly the kind of question that Kenyan scientists at the IGAD Climate Prediction and Applications Centre in Nairobi are working on right now. You are thinking like researchers.' WAIT TIME: After asking which new question excites them most, wait 7 seconds to allow genuine reflection before taking responses.	enterprise.	the connection between their DQB answers and specific lessons or evidence sources.
Model Building Phase 📹 **VIDEO:** Video: Molecules and Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group has 3 minutes to do a final check of their annotated cumulative model from Component A of the Final Explanation Sheet, adding any details they noticed were missing during the DQB closure phase. Then one representative from each group holds up their model and gives a 60-second verbal summary: 'Our model shows...' The class listens and after each presentation has 30 seconds to identify one thing this group included that their own model did not. After all groups have presented (approximately 6 minutes total for a class of 5 groups), the teacher facilitates a 2-minute consensus-building discussion: 'Looking at all the models together — what are the three things every model agrees on? And where do we still see honest differences between models?' The lesson closes with the teacher reading the Embu farmer testimony one final time, then asking students to write one sentence in their exercise books: 'I can now explain the farmer's	Give groups 3 minutes for final model revision. Then say: 'Each group will have 60 seconds to present their model. Listeners: your job is to find one thing this group included that your group missed. Be generous — celebrate what others figured out that you did not.' Facilitate presentations briskly, keeping each group to 60 seconds. After all presentations, say: 'Without looking at your notes — what are the three things every single model in this room agrees on? Call them out.' Accept contributions and write three consensus statements on the board. Then ask: 'Where do our models honestly differ? That is not a failure — that is science.' After consensus discussion, read the Embu farmer testimony slowly one final time. Then say: 'In your exercise book, write one sentence starting with: I can now explain the farmer experience using physics because... You have 90 seconds. Then share it with the person next to you.' Close the lesson by saying: 'Seven lessons ago, you	The 60-second group model presentation combines accountability (every group must synthesise) with peer learning (every student actively compares). The consensus-building discussion models the scientific practice of identifying agreed explanations versus ongoing uncertainties — directly mirroring how the IPCC synthesises climate science from multiple research groups. The final one-sentence writing task is a low-stakes but high-meaning closure: it requires students to personally connect physics to lived human experience, fulfilling the affective and citizenship goals of the unit. The re-reading of the testimony creates emotional bookending with Lesson 1, honouring the human stakes of the science.	The teacher listens to the final one-sentence statements shared with partners and notes: (1) whether students use physical language (infrared, greenhouse gas, energy balance, water cycle) or remain at the level of everyday language (it gets hot); (2) whether students make the connection between global atmospheric physics and the specific local experience of a Kenyan farmer; (3) whether any students express a sense of personal agency or responsibility — these responses are especially worth celebrating aloud as they demonstrate the full CBE competency profile. The Component B written explanations collected at the end of the Explain phase serve as the formal summative assessment for this lesson and for the unit.

	experience using physics because...' Students share this sentence aloud with a partner before the bell.	saw a glacier disappearing and a farmer struggling, and you did not know why. Today, you are physicists who can explain it. That matters — because Kenya needs people who can think like this.' WAIT TIME: After the final reading of the testimony and before asking students to write, wait 6 seconds in silence to allow emotional and intellectual resonance.		

D. TEACHER REFLECTION

After the lesson, consider: Did students demonstrate durable understanding of the infrared absorption mechanism, or did many revert to surface descriptions like 'CO2 makes it hot'? Were students able to articulate the specific connection between atmospheric warming and rainfall disruption in Embu — the water cycle and shifting rainfall patterns — or did they only explain glacier retreat? Did the DQB closure reveal genuine metacognitive awareness of what is known versus uncertain, or did students treat all questions as fully answered? Were the red cards intellectually ambitious? Did the final one-sentence statements use physical language precisely? If students struggled to connect the physics mechanism to the local Kenyan consequences, consider whether future iterations should include more explicit causal chain work in Lesson 4. If students struggled with the written explanation, consider providing a sentence frame scaffold in Component B for students who need it: 'Human activities such as [example] release [greenhouse gas], which absorbs [type of radiation] in the atmosphere. This causes [effect], which in Kenya leads to [local consequence]. Kenyans are responding by [action].' Celebrate the learning journey with students — this unit asks them to think at the scale of the entire atmosphere and connect it to the life of one farmer in Embu, and that is genuinely difficult and genuinely important.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed the original anchoring phenomenon images of the Lewis Glacier (1934 versus today) and re-read the Embu farmer testimony. We observed our own cumulative models and those of our peers, noticing where they agreed and where they differed. We observed the Driving Question Board from Lesson 1 and saw how many of our original questions could now be answered.
What did I learn?	We learned that a single connected causal chain — from human greenhouse gas emissions through infrared trapping to surface warming to glacier retreat and water cycle disruption — explains both the disappearing Lewis Glacier and the unpredictable long rains in Embu. We learned that our models across the class share a strong consensus on the core mechanism while retaining honest differences on details, which reflects how real climate science works. We learned that Kenyan actions — from Olkaria Geothermal to household energy-efficient jikos — represent genuine responses within this causal chain.
How does this explain the phenomenon?	We explained, in writing and in visual models, the full physics of why Mount Kenya's glaciers are retreating and why Embu farmers can no longer predict the long rains. Our explanation connects solar radiation input, greenhouse gas absorption of outgoing infrared radiation, enhanced warming from human-caused emission increases, glacier mass loss, disruption of the East African water cycle, and Kenya-specific mitigation and adaptation responses — forming a complete, evidence-based scientific account of the anchoring phenomenon first posed in Lesson 1.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.